Thesis draft

Md. Mezbah Uddin

Contents

Chapter 1

Introduction

The central theme of this thesis is the interplay between quantum field theory and cosmology during the early universe. The key observation is this: if matter is described by a quantum field with a nonvanishing potential, there is a natural mechanism for generating an effective cosmological constant in the early universe, which drives a phase of exponential expansion known as inflation. This in turn provides an elegant explanation for the homogeneity and flatness of the observed universe.

We work within the framework of grand unified theories (GUTs), where gravity is described classically by a Friedmann-Robertson-Walker (FRW) metric and matter is described by quantum field theory. At $T = 0$, the gauge symmetry of the GUT is spontaneously broken by a nonzero vacuum expectation value of a scalar Higgs field. At high temperature the symmetry is restored, but at the cost of generating a large cosmological constant. As the universe cools, it passes through one or more phase transitions.

The reason for describing matter using quantum fields is well-motivated: for times earlier than 10^{-6} s after the Big Bang, the thermal energy of particles exceeds 1 GeV, and at these energy scales particle interactions are correctly described by QFT, as confirmed by particle accelerator experiments. An ideal gas approximation is simply inadequate at such temperatures.

The central tool of this thesis is the *effective potential* $V_{\text{eff}}(\bar{\varphi})$. To understand why it is the right object, consider a scalar field φ coupled to other fields A_μ. The classical potential $V(\varphi)$ gives the energy of a homogeneous stationary state with $\varphi(x) = \varphi$ and $A_\mu = 0$. The effective potential $V_{\text{eff}}(\bar{\varphi})$ is the minimum expectation value of the energy density in the class of all homogeneous stationary states with expectation value $\bar{\varphi}$. This is the main reason it is of interest: the value of $\bar{\varphi}$ in the quantum state at any given time minimizes $V_{\text{eff}}(\bar{\varphi})$ at that time.

A large positive potential $V(\phi)$ in some range of field values, with a quantum state that is homogeneous and stationary in that range, causes the potential contribution to the energy-momentum tensor to dominate, acting as an effective cosmological constant in Einstein's equations. Finding the state of the quantum field as the universe evolves therefore reduces to finding the minimum of the effective potential.

The effective potential also governs the equation of motion of the expectation value $\bar{\varphi}$,

$$\frac{\partial^2 \bar{\varphi}}{\partial t^2} - \nabla^2 \bar{\varphi} = -\frac{\partial}{\partial \bar{\varphi}} V_{\text{eff}}(\bar{\varphi}), \tag{1.1}$$

which is the leading contribution in an expansion about a homogeneous and stationary state. The effective potential thus replaces the classical potential in the quantum equation of motion, as we will show explicitly.

Two reasons motivate the use of grand unified gauge field theories in this context. First, particle accelerator experiments suggest that fundamental interactions are described by gauge theories at high energies. Second, the cosmological constant Λ vanishes today, $\Lambda/m_{\mathrm{Pl}}^2 < 10^{-110}$, so for Λ to have been large in the early universe, a phase transition from $\langle V(\phi)\rangle \neq 0$ to $\langle V(\phi)\rangle = 0$ must have occurred. GUTs have such phase transitions built in.

The temperature dependence of the effective potential is crucial here. Since the effective action is the free energy of the quantum state, the effective potential is the free energy density for homogeneous states, and statistical mechanics tells us that the free energy is temperature dependent. Concretely, finite-temperature corrections generate a mass term proportional to $\varphi^2 T^2$, which restores the symmetry at high temperature: even if $\varphi = 0$ is not the minimum at $T = 0$, it becomes so at sufficiently high T. The high-T and low-T ground states are therefore different.

In GUT models with a Higgs mechanism, the gauge symmetry is broken by a nonvanishing VEV of φ. Finite-temperature corrections to the effective potential restore the symmetry above a critical temperature T_c, set by the scale of symmetry breaking. As the universe cools below T_c, a new absolute minimum of $V_{\mathrm{eff}}(\varphi)$ appears at $\varphi = \sigma \neq 0$ and a phase transition occurs. Since Λ must be fine-tuned to zero today, $V_{\mathrm{eff}}(\sigma) = 0$, it follows that $V_{\mathrm{eff}}(0) > 0$ in the high-T phase. GUTs with Higgs symmetry breaking therefore automatically generate a period of inflation preceding the phase transition.

In the semiclassical approximation, the decay of the false vacuum proceeds via *bubble nucleation*: the true stable phase, with vanishing Λ, nucleates as a bubble and expands at the speed of light into the surrounding sea of false vacuum. The analysis of this process is one of the main topics of this thesis.

In curved spacetime, new effects arise. Unlike flat spacetime, where the Poincaré-invariant Minkowski vacuum is agreed upon by all inertial observers, general relativity is diffeomorphism invariant and no frame is globally preferred. As a consequence, there is no unique vacuum state in a general curved spacetime. The most celebrated manifestation of this is Hawking radiation: an observer with a particle detector in certain spacetimes will, at late times, measure a nonvanishing thermal flux of particles in a state that was initially empty. We discuss these effects in the final part of the thesis.

The thesis is organized as follows. Part I develops the effective action formalism at zero temperature, building from the generating functionals to the one-loop effective potential and its application to gauge theories. Part II extends this to finite temperature using the imaginary-time formalism. Part III treats cosmological phase transitions, including the thermal history of the early universe and the parameters characterizing a first-order transition. Part IV analyzes false vacuum decay and bubble nucleation via the bounce solution. Part V discusses quantum field theory in curved spacetime, culminating in Hawking radiation.

Part I

Effective Action and Effective Potential

Chapter 2

General Formalism of the Effective Action

The central object of this thesis is the *effective action* $\Gamma[\varphi_c]$, a functional that encodes the full quantum dynamics of a field theory in a single object. Before defining it, we build up to it systematically: starting from the generating functional $Z[J]$ for all Green functions, passing through the connected generating functional $W[J]$, and finally arriving at $\Gamma[\varphi_c]$ via a Legendre transform. The payoff is that the equation of motion derived from Γ governs the quantum-corrected vacuum structure of the theory, making it the natural tool for studying spontaneous symmetry breaking and phase transitions.

2.1 Generating Functionals

Rather than computing Green functions diagram by diagram, all of them, full, connected, and one-particle-irreducible, can be extracted from a small number of functionals by simple functional differentiation.

Coupling the field $\varphi(x)$ linearly to an arbitrary c-number source $J(x)$,

$$\mathcal{L}(\varphi, \partial_\mu \varphi) \;\longrightarrow\; \mathcal{L} + J(x)\,\varphi(x), \tag{2.1}$$

the generating functional for the full Green functions is defined as the vacuum-to-vacuum transition amplitude in the presence of this source,

$$Z[J] = \langle 0 | T \exp\!\left(i \int J(x)\,\varphi(x)\, d^4x \right) | 0 \rangle, \tag{2.2}$$

where $|0\rangle$ is the physical vacuum and $\varphi(x)$ is the Heisenberg-picture field, both defined at $J = 0$. Expanding in powers of J makes the content of this definition explicit,

$$Z[J] = \sum_{n=0}^{\infty} \frac{i^n}{n!} \int d^4x_1 \cdots d^4x_n \; J(x_1) \cdots J(x_n)\; G_n(x_1, \ldots, x_n), \tag{2.3}$$

where $G_n(x_1, \ldots, x_n) \equiv \langle 0 | T[\varphi(x_1) \cdots \varphi(x_n)] | 0 \rangle$ is the full n-point Green function. Each G_n is recovered by n functional derivatives of $Z[J]$ with respect to J, evaluated at $J = 0$. Equivalently, $Z[J]$ admits a path-integral representation,

$$Z[J] = N \int [\mathcal{D}\varphi]\; \exp\!\left(i \int d^4x \left[\mathcal{L}[\varphi(x)] + J(x)\,\varphi(x) \right] \right), \tag{2.4}$$

where N is fixed by $Z[0] = 1$. The two representations (2.2) and (2.4) describe the same object and will be used interchangeably throughout.

The full Green functions G_n contain disconnected contributions, products of lower-point functions, that carry no new dynamical information. The *connected* Green functions G_n^c, which isolate only the irreducible correlations, are generated by passing to the logarithm of $Z[J]$,

$$iW[J] = \ln Z[J], \tag{2.5}$$

the proof that this logarithm precisely removes all disconnected pieces is given in Appendix A. Since $Z[J]$ is the vacuum-to-vacuum amplitude, Eq. (2.5) means

$$e^{iW[J]} = \langle 0^+ | 0^- \rangle_J, \tag{2.6}$$

where $|0^\pm\rangle$ are the vacuum states in the far future and far past, so $W[J]$ is the *vacuum persistence amplitude* in the presence of the external source. This interpretation will be central when we discuss the metastability of the false vacuum in Chapter 15.

The first functional derivative of $W[J]$ defines the *classical field*,

$$\varphi_c(x) \equiv \frac{\delta W}{\delta J(x)} = \left[\frac{\langle 0^+ | \varphi(x) | 0^- \rangle}{\langle 0^+ | 0^- \rangle} \right]_J, \tag{2.7}$$

the source-dependent vacuum expectation value of $\varphi(x)$, which will serve as the argument of the effective action.

2.2 The Effective Action and the Effective Potential

Given $W[J]$ and the classical field φ_c via (2.7), we trade the source J for φ_c as the fundamental variable via the Legendre transform,

$$\Gamma[\varphi_c] = W[J] - \int d^4x \, J(x) \, \varphi_c(x), \tag{2.8}$$

where J on the right-hand side is understood as a functional of φ_c obtained by inverting (2.7). Differentiating with respect to φ_c and using (2.7) gives

$$\frac{\delta \Gamma}{\delta \varphi_c(x)} = -J(x). \tag{2.9}$$

In the physical situation $J = 0$, this becomes the quantum equation of motion,

$$\frac{\delta \Gamma}{\delta \varphi_c(x)} = 0, \tag{2.10}$$

the exact analogue of $\delta S / \delta \varphi = 0$ for the classical action, but now with all quantum corrections included. The vacua of the quantum theory are precisely the solutions to (2.10).

Expanding $\Gamma[\varphi_c]$ in powers of φ_c,

$$\Gamma[\varphi_c] = \sum_{n=0}^{\infty} \frac{1}{n!} \int d^4x_1 \cdots d^4x_n \, \Gamma^{(n)}(x_1, \ldots, x_n) \, \varphi_c(x_1) \cdots \varphi_c(x_n), \tag{2.11}$$

the coefficients $\Gamma^{(n)}$ are the one-particle-irreducible (1PI) Green functions, or proper vertices, the sum of all Feynman diagrams with n external lines that cannot be disconnected by cutting a single internal propagator, with no external propagators attached.

For a translationally invariant configuration $\varphi_c = \text{const}$, the effective action takes the form $\Gamma = -\Omega\, V(\varphi_c)$, where Ω is the four-volume. More generally, for slowly-varying fields, Γ admits a derivative expansion,

$$\Gamma[\varphi_c] = \int d^4x \left[-V(\varphi_c) + \frac{1}{2}(\partial_\mu \varphi_c)^2\, Z(\varphi_c) + \cdots \right], \tag{2.12}$$

where $V(\varphi_c)$ is the *effective potential*, an ordinary function rather than a functional. Comparing with (2.11), the n-th derivative of V equals the sum of all 1PI diagrams with n external legs at zero external momenta, so the effective potential encodes the full quantum-corrected vacuum structure of the theory. The physical mass and coupling are read off from it via

$$\mu^2 = \left.\frac{d^2V}{d\varphi_c^2}\right|_{\varphi_c=0}, \qquad \lambda = \left.\frac{d^4V}{d\varphi_c^4}\right|_{\varphi_c=0}. \tag{2.13}$$

2.3 Spontaneous Symmetry Breaking

The effective potential is the natural language for spontaneous symmetry breaking. Suppose $\mathcal{L}$ has a discrete internal symmetry, e.g. $\varphi \to -\varphi$. This symmetry is spontaneously broken when the quantum equation of motion (2.10) admits a solution with a nonzero, spatially uniform vacuum expectation value,

$$\left.\frac{dV}{d\varphi_c}\right|_{\varphi_c=\langle\varphi\rangle} = 0, \qquad \langle\varphi\rangle \neq 0, \tag{2.14}$$

even in the absence of any external source. Stability requires this stationary point to be a minimum of V.

Two phases are possible. In the symmetric phase, $\langle\varphi\rangle = 0$ and the $\varphi \to -\varphi$ symmetry is intact. In the broken phase, $\langle\varphi\rangle = v \neq 0$ and the symmetry is hidden. The physical spectrum in the broken phase is described by the shifted field $\varphi' = \varphi - v$, and the renormalization conditions (2.13) are displaced to the broken minimum,

$$\mu^2 = \left.\frac{d^2V}{d\varphi_c^2}\right|_{\varphi_c=v}, \qquad \lambda = \left.\frac{d^4V}{d\varphi_c^4}\right|_{\varphi_c=v}. \tag{2.15}$$

With the formalism in place, we turn in the next chapter to an explicit one-loop computation of the effective potential, first in a simple scalar theory and then in the physically relevant case of scalar electrodynamics.

Chapter 3

Physical Meaning of the Effective Action and Effective Potential

We have defined $\Gamma[\varphi_c]$ and $V_{\text{eff}}(\bar{\varphi})$ formally in the previous chapter. Here we give them concrete physical meaning.

3.1 The Effective Potential as a Generator of 1PI Graphs

For a constant classical field $\bar{\varphi}$, inserting the Fourier representation of $\Gamma^{(n)}$ into the expansion (2.11) gives

$$\Gamma(\bar{\varphi}) = \sum_{n=2}^{\infty} \frac{1}{n!}\,\bar{\varphi}^n \int \prod_{i=1}^{n} d^4x_i\, \Gamma^{(n)}(x_1,\ldots,x_n) \tag{3.1}$$

$$= \sum_{n=2}^{\infty} \frac{1}{n!}\,\bar{\varphi}^n \int \prod_{i=1}^{n} \left(d^4x_i\, \frac{d^4k_i}{(2\pi)^4}\, e^{ik_ix_i} \right) \tilde{\Gamma}^{(n)}(k_1,\ldots,k_n)\cdot (2\pi)^4\,\delta^{(4)}\!\left(\sum_{i=1}^{n} k_i\right), \tag{3.2}$$

where $\int d^4x\, e^{ikx} = (2\pi)^4\delta^{(4)}(k)$ was used. Setting $k=0$ gives $\int d^4x = (2\pi)^4\delta^{(4)}(0)$, so that

$$\Gamma(\bar{\varphi}) = \sum_{n=2}^{\infty} \frac{1}{n!}\,\bar{\varphi}^n \int d^4x\, \tilde{\Gamma}^{(n)}(0,\ldots,0). \tag{3.3}$$

Comparing this with $\Gamma(\bar{\varphi}) = \int d^4x\,[-V_{\text{eff}}(\bar{\varphi})]$ gives

$$V_{\text{eff}}(\bar{\varphi}) = -\sum_{n=2}^{\infty} \frac{1}{n!}\,\bar{\varphi}^n\, \tilde{\Gamma}^{(n)}(0,\ldots,0). \tag{3.4}$$

Thus $V_{\text{eff}}(\bar{\varphi})$ is the generating functional for 1PI graphs at vanishing external momenta, computed by summing all such graphs and inserting a factor $\bar{\varphi}$ for each external line.

To lowest order in $\hbar$, only tree-level graphs contribute. For a Lagrangian of the form

$$\mathcal{L}(\varphi, \partial_\mu\varphi) = \frac{1}{2}\,\partial_\mu\varphi\,\partial^\mu\varphi - U(\varphi), \tag{3.5}$$

each term $g_i\varphi^i/i!$ in $U(\varphi)$ contributes exactly one tree-level 1PI graph, since all tree graphs with more than one vertex are not 1PI. Each term gives a contribution $-g_i\bar{\varphi}^i$, where $-g_i$

is the vertex factor and $\bar{\varphi}^i$ comes from the i external lines. The minus signs cancel and one recovers

$$V_{\text{eff}}(\bar{\varphi}) = U(\bar{\varphi}) + \mathcal{O}(\hbar), \tag{3.6}$$

confirming that the classical potential is the leading-order approximation to the effective potential.

3.2 The Effective Potential as a Minimum Energy Density

A deeper interpretation follows from the variational principle.

Theorem 1. *$V_{\text{eff}}(\bar{\varphi})$ is the minimum of the energy density expectation value over all normalized states $|a\rangle$ satisfying $\langle a|\varphi|a\rangle = \bar{\varphi}$. That is,*

$$V_{\mathit{eff}}(\bar{\varphi}) = \langle a|\mathcal{H}|a\rangle, \tag{3.7}$$

for $|a\rangle$ which obeys $\delta\langle a|\mathcal{H}|a\rangle = 0$ subject to the constraints $\langle a|a\rangle = 1$, $\langle a|\varphi|a\rangle = \bar{\varphi}$.

Proof. We first solve the analogous problem in quantum mechanics, then translate to field theory. In QM the problem is to find a state $|a\rangle$ such that $\langle a|H|a\rangle$ is stationary subject to $\langle a|a\rangle = 1$ and $\langle a|A|a\rangle = A_c$ for some self-adjoint operator A. Introducing Lagrange multipliers E and J for each constraint, the problem becomes

$$\delta\langle a|(H - E - JA)|a\rangle = 0 \quad \text{(unconstrained)} \tag{3.8}$$

$$\langle a|a\rangle = 1 \tag{3.9}$$

$$\langle a|A|a\rangle = A_c. \tag{3.10}$$

Since H is self-adjoint, this yields

$$(H - E - JA)|a\rangle = 0, \tag{3.11}$$

with solution $|a\rangle = |a(E, J)\rangle$. Using $\langle a|a\rangle = 1$, E is solved as a function of J, i.e. $E = E(J)$, and $\langle a|A|a\rangle = A_c$ gives $J = J(A_c)$. Thus

$$(H - JA)|a(J)\rangle = E(J)|a(J)\rangle, \tag{3.12}$$

where $|a(J)\rangle$ is a normalized eigenstate of $(H - JA)$. Taking the expectation value with $\langle a(J)|$ and differentiating with respect to J,

$$\begin{aligned}\left(\frac{\delta}{\delta J}\langle a(J)|\right)\underbrace{(H - E(J) - JA)|a(J)\rangle}_{0} + \langle a(J)|\frac{\delta}{\delta J}(H - E(J) - JA)|a(J)\rangle \\ + \langle a(J)|(H - E(J) - JA)\underbrace{\frac{\delta}{\delta J}|a(J)\rangle}_{0} = 0,\end{aligned} \tag{3.13}$$

which gives

$$-\frac{\delta E}{\delta J} - \underbrace{\langle a(J)|A|a(J)\rangle}_{A_c} = 0 \qquad \Rightarrow \qquad \frac{\delta E}{\delta J} = -A_c. \tag{3.14}$$

Substituting back into the expectation value of the eigenvalue equation,

$$\langle a(J)|H|a(J)\rangle = E(J) - J\frac{\delta E}{\delta J}, \tag{3.15}$$

which is the energy of the ground state of H in the space of states satisfying $\langle a(J)|A|a(J)\rangle = A_c$.

We now translate this to field theory. In field theory the only normalizable eigenstate of a Hamiltonian is the vacuum, so $|a(J)\rangle$ is identified with the vacuum of the sourced theory. Adding a source $J(x)\varphi(x)$ to the Hamiltonian density, adiabatically switched on at $t = 0$ and off at $t = T$ over spatial volume V, the adiabatic theorem gives

$$e^{iW(J)} = \langle 0^+|0^-\rangle = \exp\left[-iVT\,\varepsilon(J)\right], \tag{3.16}$$

so $-W(J)$ is the ground state action of the sourced theory. The Legendre transform structure,

$$\Gamma(\bar{\varphi}) = W(J) - J\frac{\delta W}{\delta J}, \qquad \frac{\delta W}{\delta J} = \bar{\varphi}, \tag{3.17}$$

is formally identical to the quantum-mechanical result above, identifying $-\Gamma(\bar{\varphi})$ as the ground state action of the sourceless theory in the sector with $\langle a|\varphi|a\rangle = \bar{\varphi}$. That is,

$$-\Gamma(\bar{\varphi}) = \langle a|H|a\rangle, \tag{3.18}$$

$$-\frac{\Gamma(\bar{\varphi})}{VT} = \frac{\langle a|H|a\rangle}{VT}, \tag{3.19}$$

and therefore

$$V_{\text{eff}}(\bar{\varphi}) = \langle a|\mathcal{H}|a\rangle \tag{3.20}$$

for the lowest-energy state satisfying $\langle a|a\rangle = 1$ and $\langle a|\varphi|a\rangle = \bar{\varphi}$. □

This result has an important consequence: since $V_{\text{eff}}(\bar{\varphi})$ is defined as a minimum of energy over a convex set of states, it is convex. Its minima therefore correspond to the stable equilibrium configurations of the quantum field theory, a fact that will be central to the analysis of phase transitions in later chapters.

3.3 Variational Definition of the Full Effective Action

The variational interpretation of V_{eff} established above can be generalized to the full effective action.

Theorem 2. $\Gamma(\varphi)$ *is the stationary value of*

$$\Gamma(\varphi) = \int_{-\infty}^{\infty} dt\,\langle\psi_-, t|\, i\partial_t - H\,|\psi_+, t\rangle \tag{3.21}$$

subject to the constraints

$$\langle\psi_-, t|\Phi(\mathbf{x})|\psi_+, t\rangle = \varphi(t, \mathbf{x}), \qquad \langle\psi_-, t|\psi_+, t\rangle = 1, \qquad \lim_{t\to\pm\infty} |\psi_\pm, t\rangle = |0\rangle. \tag{3.22}$$

Proof. The proof follows the same structure as Theorem 2. Introducing Lagrange multipliers $J(t,\mathbf{x})$ and $w(t)$ for the two constraints,

$$\Gamma_m(\varphi) = \int dt\, \langle\psi_-,t|i\partial_t - H|\psi_+,t\rangle + \int d^4x\, J(t,\mathbf{x})\,(\langle\psi_-,t|\Phi|\psi_+,t\rangle - \varphi(t,\mathbf{x})) \\ - w(t)\,(\langle\psi_-,t|\psi_+,t\rangle - 1)\,. \quad (3.23)$$

Varying with respect to $\langle\psi_-,t|$, the last term drops out and since $\langle\delta\psi_-,t|$ is arbitrary,

$$\boxed{\left(i\partial_t - H + \int d^3x\, J(t,\mathbf{x})\,\Phi(\mathbf{x})\right)|\psi_+,t\rangle = w(t)|\psi_+,t\rangle.} \quad (3.24)$$

Varying with respect to $|\psi_+,t\rangle$ and integrating by parts,

$$\int dt\, \langle\psi_-,t|i\partial_t|\delta\psi_+,t\rangle = [\langle\psi_-,t|\delta\psi_+,t\rangle]_{-\infty}^{\infty} - \int dt\, i\,(\partial_t\langle\psi_-,t|)\,|\delta\psi_+,t\rangle, \quad (3.25)$$

where the boundary term vanishes because $\lim_{t\to\pm\infty}|\psi_\pm,t\rangle = |0\rangle$. Hence

$$\int dt\, \langle\psi_-,t|i\partial_t|\delta\psi_+,t\rangle = -\int dt\, i\,(\partial_t\langle\psi_-,t|)\,|\delta\psi_+,t\rangle. \quad (3.26)$$

Since $|\delta\psi_+,t\rangle$ is arbitrary,

$$\delta_{|\psi_+,t\rangle}\Gamma_m(\varphi) = \int dt\, \langle\psi_-,t|\left(-i\overleftarrow{\partial_t} + H + \int d^3x\, J(\mathbf{x})\,\Phi(\mathbf{x}) + w(t)\right)|\delta\psi_+,t\rangle = 0, \quad (3.27)$$

and taking the Hermitian conjugate,

$$\boxed{\left[i\partial_t - H + \int d^3x\, J(\mathbf{x})\,\Phi(\mathbf{x})\right]|\psi_-,t\rangle = w^*(t)|\psi_-,t\rangle.} \quad (3.28)$$

Defining the rescaled states

$$|+,t\rangle = e^{i\int_{-\infty}^{t} dt'\, w(t')}\,|\psi_+,t\rangle, \quad (3.29)$$
$$|-,t\rangle = e^{-i\int_{t}^{\infty} dt'\, w^*(t')}\,|\psi_-,t\rangle, \quad (3.30)$$

these are asymptotic ground states satisfying the sourced Schrödinger equation with no inhomogeneous term. Their overlap gives

$$\langle -,t|+,t\rangle = e^{iW(J)} = \exp\left[i\int_{-\infty}^{\infty} dt\, w(t)\right], \quad (3.31)$$

from which

$$W(J) = \int_{-\infty}^{\infty} dt\, \langle\psi_-,t|\, i\partial_t - H + \int d^3x\, J\cdot\Phi\,|\psi_+,t\rangle. \quad (3.32)$$

As shown in Appendix D, $\delta W(J)/\delta J = \varphi$, so the Legendre transform gives

$$W(J) - \int d^4x\, J\varphi = \Gamma(\varphi) = \int_{-\infty}^{\infty} dt\, \langle\psi_-,t|\, i\partial_t - H\,|\psi_+,t\rangle, \quad (3.33)$$

which shows that $\Gamma(\varphi)$ is indeed the stationary value. □

The equation (2.9) has a further physical interpretation. When the source vanishes, $\delta\Gamma/\delta\bar{\varphi} = 0$ becomes the exact quantum equation of motion for $\varphi(\mathbf{x}, t)$. Expanding Γ in powers of momentum about vanishing external momenta,

$$\Gamma(\varphi) = \int d^4x \left[-V_{\text{eff}}(\varphi) + \frac{1}{2} \left(\partial_\mu \varphi \, \partial^\mu \varphi \right) Z(\varphi) + \cdots \right], \tag{3.34}$$

and renormalizing the field so that $Z(\varphi) = 1$, the equation of motion becomes

$$\Box \, \varphi = -V'_{\text{eff}}(\varphi). \tag{3.35}$$

In the static limit this is precisely the classical Klein-Gordon equation, but with the classical potential replaced by the effective potential. Thus quantum corrections to the dynamics are fully captured by V_{eff}, which is yet another reason it is the central object of study.

Chapter 4

Toy Model: One-Loop Effective Potential and Renormalization

We illustrate the general formalism of the previous chapter with an explicit computation in the simplest setting: a single real scalar field with Lagrangian

$$\mathcal{L}(x) = \frac{1}{2}\,\partial_\mu\phi(x)\,\partial^\mu\phi(x) - V(\phi(x)), \tag{4.1}$$

and classical potential

$$V(\phi) = \frac{\alpha_0}{4!}\,\phi^4 - \frac{m_0^2}{2}\,\phi^2. \tag{4.2}$$

We work in units $\hbar = c = 1$ throughout.

4.1 The Classical Field and the Effective Action

The classical minimum of $V(\phi)$ sits at some ϕ_0, but the vacuum expectation value $\langle\phi\rangle$ of the quantum field need not coincide with it. It is defined by

$$\langle\phi\rangle \equiv \lim_{J\to 0} \frac{\langle 0^+|\phi_{\rm op}|0^-\rangle_J}{\langle 0^+|0^-\rangle_J}, \tag{4.3}$$

where the generating functional $W[J]$ encodes the full quantum theory via

$$\exp\frac{i}{\hbar}W[J] = \mathcal{N}\int(\mathcal{D}\phi)\exp\frac{i}{\hbar}\left\{S[\phi] + (J,\phi)\right\}, \tag{4.4}$$

with $S[\phi] = \int d^4x\,\mathcal{L}(x)$ and $(J,\phi) = \int d^4x\,J(x)\,\phi(x)$. The classical field, the VEV in the presence of an external source $J(x)$, is

$$\phi_c(x) \equiv \frac{\langle 0^+|\phi_{\rm op}(x)|0^-\rangle_J}{\langle 0^+|0^-\rangle_J} = \frac{\delta W[J]}{\delta J(x)}, \tag{4.5}$$

and $\langle\phi\rangle$ is recovered as $\lim_{J\to 0}\phi_c(x)$. The loop expansion is an expansion in powers of $\hbar\,c\,\alpha$, but it performs an infinite resummation of Feynman graphs rather than a perturbative truncation.

The natural question is: given a desired profile ϕ_c, what source J produces it? To answer this, we trade J for ϕ_c as the independent variable via the Legendre transform,

$$\Gamma[\phi_c] \equiv W[J] - (J,\phi_c), \tag{4.6}$$

where J is eliminated in favour of ϕ_c by inverting (4.5). The functional $\Gamma[\phi_c]$ is the effective action, so named because it is a functional of the classical field ϕ_c and thus plays the same role for the quantum theory as $S[\phi]$ does classically. Differentiating (4.6) and using (4.5),

$$\begin{aligned}\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)} &= \frac{\delta W[J]}{\delta\phi_c(x)} - J(x) - \int d^4y\,\frac{\delta J(y)}{\delta\phi_c(x)}\,\phi_c(y),\\ \frac{\delta W[J]}{\delta\phi_c(x)} &= \int d^4y\,\frac{\delta W[J]}{\delta J(y)}\,\frac{\delta J(y)}{\delta\phi_c(x)} = \int d^4y\,\frac{\delta J(y)}{\delta\phi_c(x)}\,\phi_c(y),\end{aligned} \tag{4.7}$$

so that

$$\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)} = -J(x). \tag{4.8}$$

At $J=0$, translational invariance forces ϕ_c to be a constant, so $\langle\phi\rangle$ is the root of

$$\left.\frac{d\Gamma[\phi_c]}{d\phi_c}\right|_{\phi_c=\langle\phi\rangle} = 0. \tag{4.9}$$

4.2 The Effective Action as a Generator of 1PI Green's Functions

Expanding $\Gamma[\phi_c]$ in powers of ϕ_c,

$$\Gamma[\phi_c] = \sum_{n=0}^{\infty}\frac{1}{n!}\int d^4x_1\cdots d^4x_n\,\Gamma_n(x_1,\ldots,x_n)\,\phi_c(x_1)\cdots\phi_c(x_n), \tag{4.10}$$

we claim that Γ_n is the n-point 1PI Green function, the sum of all connected 1PI Feynman graphs with n external legs. To see this, consider the functional integral

$$\mathcal{N}\int(\mathcal{D}\phi)\exp\frac{i}{\hbar}\left\{\Gamma[\phi]+(J,\phi)\right\}. \tag{4.11}$$

In the limit $\hbar\to 0$ the integrand is dominated by its saddle point, which occurs at $\phi=\phi_c$ since (4.8) is precisely the saddle-point condition. Thus

$$\mathcal{N}\int(\mathcal{D}\phi)\exp\frac{i}{\hbar}\left\{\Gamma[\phi]+(J,\phi)\right\} \xrightarrow{\hbar\to 0} \mathcal{N}\exp\frac{i}{\hbar}\left\{\Gamma[\phi_c]+(J,\phi_c)\right\}, \tag{4.12}$$

and by (4.6) the right-hand side equals $\exp\frac{i}{\hbar}W[J]$, so

$$\exp\frac{i}{\hbar}W[J] \xrightarrow{\hbar\to 0} \mathcal{N}\int(\mathcal{D}\phi)\exp\frac{i}{\hbar}\left\{\Gamma[\phi]+(J,\phi)\right\}. \tag{4.13}$$

In this limit the right-hand side is the sum of all tree graphs of a theory whose action is $\Gamma[\phi]$, with vertices given by the Γ_n of (4.10). Since $W[J]$ generates all connected Green functions of the original theory, and any connected graph decomposes into 1PI components, it follows that Γ_n is a 1PI Green function of the original theory. $\square$

4.3 The Effective Potential

Alternatively, $\Gamma[\phi_c]$ may be expanded in derivatives of ϕ_c,

$$\Gamma[\phi_c] = \int d^4x \left[-U(\phi_c(x)) + \frac{1}{2}\, \partial_\mu \phi_c(x)\, \partial^\mu \phi_c(x)\, Z(\phi_c(x)) + \cdots \right], \tag{4.14}$$

where $U(\phi_c)$ is the effective potential. For a constant configuration $\phi_c(x) = \rho$ this reduces to

$$\Gamma[\rho] = -\Omega\, U(\rho), \tag{4.15}$$

where Ω is the total spacetime volume. To connect $U(\rho)$ to the 1PI Green functions, we pass to momentum space. The Fourier transforms of ϕ_c and Γ_n are defined by

$$\tilde{\phi}_c(k) = \int d^4x\, e^{ik\cdot x}\, \phi_c(x), \tag{4.16}$$

$$\phi_c(x) = \int \frac{d^4k}{(2\pi)^4}\, e^{-ik\cdot x}\, \tilde{\phi}_c(k), \tag{4.17}$$

$$\tilde{\Gamma}_n(k_1, \ldots, k_n)\, (2\pi)^4\, \delta^4(k_1 + \cdots + k_n) = \int d^4x_1 \cdots d^4x_n\, \Gamma_n(x_1, \ldots, x_n)\, e^{i(k_1 x_1 + \cdots + k_n x_n)}. \tag{4.18}$$

Substituting these into (4.10),

$$\begin{aligned}
\Gamma[\phi_c] &= \sum_{n=0}^{\infty} \frac{1}{n!} \int d^4x_1 \cdots d^4x_n\, \Gamma_n(x_1, \ldots, x_n)\, \phi_c(x_1) \cdots \phi_c(x_n) \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^4k_1}{(2\pi)^4} \cdots \frac{d^4k_n}{(2\pi)^4} \int d^4x_1 \cdots d^4x_n\, \Gamma_n(x_1, \ldots, x_n)\, e^{-i(k_1 x_1 + \cdots + k_n x_n)}\, \tilde{\phi}(k_1) \cdots \tilde{\phi}(k_n) \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^4k_1}{(2\pi)^4} \cdots \frac{d^4k_n}{(2\pi)^4}\, (2\pi)^4\, \delta^4(k_1 + \cdots + k_n)\, \tilde{\Gamma}_n(k_1, \ldots, k_n)\, \tilde{\phi}(k_1) \cdots \tilde{\phi}(k_n).
\end{aligned} \tag{4.19}$$

Setting $\phi_c(x) = \rho$ gives $\tilde{\phi}_c(k) = (2\pi)^4 \delta^4(k)\, \rho$, so each momentum integral collapses,

$$\begin{aligned}
\Gamma[\rho] &= \sum_{n=0}^{\infty} \frac{1}{n!} \int \frac{d^4k_1}{(2\pi)^4} \cdots \frac{d^4k_n}{(2\pi)^4}\, (2\pi)^4\, \delta^4(k_1 + \cdots + k_n)\, \tilde{\Gamma}_n(k_1, \ldots, k_n) \\
&\qquad \cdot\, (2\pi)^4 \delta^4(k_1)\, \rho \cdots (2\pi)^4 \delta^4(k_n)\, \rho \\
&= \sum_{n=0}^{\infty} \frac{1}{n!} \underbrace{(2\pi)^4 \delta^4(0)}_{\Omega}\, \rho^n\, \tilde{\Gamma}_n(0, \ldots, 0),
\end{aligned} \tag{4.20}$$

and therefore

$$\Gamma[\rho] = \Omega \sum_{n=0}^{\infty} \frac{\rho^n}{n!}\, \tilde{\Gamma}_n(0), \tag{4.21}$$

where $\tilde{\Gamma}_n(0) \equiv \tilde{\Gamma}_n(0, \ldots, 0)$ and $\Omega = (2\pi)^4 \delta^4(0)$ is the total spacetime volume. Comparing (4.21) with $\Gamma[\rho] = -\Omega\, U(\rho)$,

$$U(\rho) = -\sum_{n=0}^{\infty} \frac{\rho^n}{n!}\, \tilde{\Gamma}_n(0), \tag{4.22}$$

which is simply the statement that the Taylor coefficients of U are the zero-momentum 1PI Green functions,

$$-\frac{d^nU(\rho)}{d\rho^n}\bigg|_{\rho=0} = \tilde{\Gamma}_n(0). \tag{4.23}$$

The effective potential is thus entirely determined by the 1PI Green functions of the original theory evaluated at vanishing external momenta. In the next section we compute it explicitly at one loop.

4.4 Spontaneous Symmetry Breaking and the Shifted Field

When spontaneous symmetry breaking occurs, it is more natural to work with the shifted field

$$\eta(x) = \phi(x) - \langle\phi\rangle, \tag{4.24}$$

so that the effective potential in terms of the shifted variable becomes

$$U(\rho - \langle\phi\rangle) = -\sum_{n=0}^{\infty} \frac{[\rho - \langle\phi\rangle]^n}{n!}\, \widetilde{\Gamma}_n^{(s)}(0), \tag{4.25}$$

where $\widetilde{\Gamma}_n^{(s)}(0)$ are the 1PI Green functions defined with respect to $\eta(x)$. The physically relevant quantities extracted from $U(\rho)$ at the minimum $\rho = \langle\phi\rangle$ are

$$\frac{dU(\rho)}{d\rho}\bigg|_{\rho=\langle\phi\rangle} = 0, \tag{4.26}$$

which determines $\langle\phi\rangle$, together with the renormalized mass and coupling constant,

$$\frac{d^2U(\rho)}{d\rho^2}\bigg|_{\rho=\langle\phi\rangle} = m^2, \qquad \frac{d^4U(\rho)}{d\rho^4}\bigg|_{\rho=\langle\phi\rangle} = \alpha. \tag{4.27}$$

Since $m^2 \geq 0$, these conditions mean that $\langle\phi\rangle$ sits at a minimum of $U(\rho)$, in direct analogy with ρ_0 being a minimum of the classical potential $V(\rho)$. The parameters m^2 and α are the free parameters of the renormalized theory.

4.5 One-Loop Effective Potential

We now compute $U(\rho)$ to one-loop order via saddle-point integration. Starting from

$$\exp\frac{i}{\hbar}W[J] = \mathcal{N}\int \mathcal{D}\phi\, \exp\frac{i}{\hbar}\left\{S[\phi] + (J,\phi)\right\}, \tag{4.28}$$

the saddle point occurs at $\phi = \phi_0$, defined by

$$\frac{\delta S[\phi]}{\delta\phi(x)}\bigg|_{\phi=\phi_0} = -J(x). \tag{4.29}$$

Here $\phi_0(x)$ is a functional of J, and as $J \to 0$ it satisfies the classical equation of motion. Expanding $S[\phi]$ about ϕ_0,

$$\begin{aligned} S[\phi_0 + \phi_c] = S[\phi_0] + \int d^4x\, \phi_c(x) \left[\frac{\delta S[\phi]}{\delta \phi}\right]_{\phi=\phi_0} \\ + \frac{1}{2}\int d^4x\, d^4y\, \phi_c(x)\, \phi_c(y) \left[\frac{\delta^2 S[\phi]}{\delta\phi(x)\,\delta\phi(y)}\right]_{\phi=\phi_0} \\ + S_2[\phi_c, \phi_0], \end{aligned} \tag{4.30}$$

where S_2 collects all higher-order terms. Introducing the propagator function

$$\langle x|\, i\Delta^{-1}[\phi_0]\, |y\rangle \equiv \left.\frac{\delta^2 S[\phi]}{\delta\phi(x)\,\delta\phi(y)}\right|_{\phi=\phi_0}, \tag{4.31}$$

the expansion becomes

$$S[\phi_0 + \phi_c] = S[\phi_0] - (J, \phi_c) + \frac{1}{2}\left(\phi_c,\, i\Delta^{-1}[\phi_0]\, \phi_c\right) + S_2[\phi_c, \phi_0]. \tag{4.32}$$

Substituting into the path integral and shifting $\phi \to \phi + \phi_0$,

$$\begin{aligned} \exp \frac{i}{\hbar} W[J] &= \mathcal{N}\int (\mathcal{D}\phi) \exp \frac{i}{\hbar}\left\{S[\phi + \phi_0] + (\phi + \phi_0, J)\right\} \\ &= \mathcal{N}\int (\mathcal{D}\phi) \exp \frac{i}{\hbar}\left\{S[\phi_0] + \frac{1}{2}(\phi,\, i\Delta^{-1}[\phi_0]\, \phi) + S_2[\phi, \phi_0] + (\phi_0, J)\right\}, \end{aligned} \tag{4.33}$$

which factors as

$$= \exp \frac{i}{\hbar}\left\{S[\phi_0] + (\phi_0, J)\right\} \cdot \left\{\mathcal{N}\int (\mathcal{D}\phi) \exp \frac{i}{2\hbar}(\phi,\, i\Delta^{-1}[\phi_0]\, \phi)\right\} \cdot \left\langle \exp \frac{i}{\hbar} S_2 \right\rangle, \tag{4.34}$$

where the Gaussian integral evaluates to $\left\{\det i\Delta^{-1}[\phi_0]\right\}^{-1/2}$ and $\langle\,\cdot\,\rangle$ denotes the functional average with respect to the Gaussian weight $\exp \frac{i}{2\hbar}(\phi,\, i\Delta^{-1}[\phi_0]\, \phi)$. Taking the logarithm,

$$W[J] = S[\phi_0] + (\phi_0, J) + \frac{i\hbar}{2} \ln \det \left\{i\Delta^{-1}[\phi_0]\right\} + W_2[J], \tag{4.35}$$

where $W_2[J] = -i\hbar \ln\langle \exp \frac{i}{\hbar} S_2\rangle$.

To determine the order of $W_2[J]$, rescale $\phi = \hbar^{1/2}\tilde{\phi}$,

$$\begin{aligned} \hbar^{-1} S_2[\phi, \phi_0] &= \hbar^{-1}\phi\phi\phi \left[\frac{\delta^3 S}{\delta\phi^3}\right]_{\phi=\phi_0} + \cdots \\ &= \hbar^{-1/2}\tilde{\phi}\tilde{\phi}\tilde{\phi} \left[\frac{\delta^3 S}{\delta\phi^3}\right]_{\phi=\phi_0} + \cdots, \end{aligned} \tag{4.36}$$

so that

$$W_2[J] = -i\hbar \ln\left[1 + \mathcal{O}(\hbar^{1/2})\right] \sim \mathcal{O}(\hbar^{3/2}). \tag{4.37}$$

Since the loop expansion of $W[J]$ contains only integer powers of $\hbar$, the fractional contribution must actually vanish and $W_2[J] \sim \mathcal{O}(\hbar^2)$. Thus the first two terms in (4.35) are the tree-level approximation and the third term is the complete one-loop correction,

$$W[J] = S[\phi_0] + (\phi_0, J) + \frac{i\hbar}{2} \ln \det \left\{i\Delta^{-1}[\phi_0]\right\} + \mathcal{O}(\hbar^2). \tag{4.38}$$

To obtain $\Gamma[\phi_c] = W[J] - (J, \phi_c)$, we express $S[\phi_0]$ as a functional of ϕ_c. At tree level $\phi_c = \phi_0$, so $\phi_c - \phi_0 \equiv \phi_1 \sim \mathcal{O}(\hbar)$,

$$\phi_c(x) \equiv \frac{\delta W[J]}{\delta J(x)} = \phi_0(x) + \phi_1(x). \tag{4.39}$$

Expanding $S[\phi_0] = S[\phi_c - \phi_1]$ about ϕ_c,

$$\begin{aligned} S[\phi_0] &= S[\phi_c] - \int d^4x\, \phi_1(x) \left\{ \frac{\delta S[\phi_0]}{\delta \phi} \right\}_{\phi=\phi_c} + \mathcal{O}(\hbar^2) \\ &= S[\phi_c] + (\phi_1, J) + \mathcal{O}(\hbar^2). \end{aligned} \tag{4.40}$$

Substituting into $\Gamma[\phi_c]$,

$$\begin{aligned} \Gamma[\phi_c] &= S[\phi_c] + \underbrace{(\phi_1, J)}_{\mathcal{O}(\hbar)} + (\phi_c, J) + \frac{i\hbar}{2} \ln\det\left\{ i\Delta^{-1}[\phi_c] \right\} + \mathcal{O}(\hbar^2) \\ &= S[\phi_c] + (\phi_c, J) + \frac{i\hbar}{2} \ln\det\left\{ i\Delta^{-1}[\phi_c] \right\} + \mathcal{O}(\hbar^2), \end{aligned} \tag{4.41}$$

where the (ϕ_1, J) term is absorbed into $\mathcal{O}(\hbar^2)$. Setting $J = 0$ and $\phi_c(x) = \rho$ (const), and using $\Gamma[\rho] = -\Omega\, U(\rho)$ and $S[\rho] = -\Omega\, V(\rho)$,

$$-\Omega\, U(\rho) = -\Omega\, V(\rho) + \frac{i\hbar}{2} \ln\det\left\{ i\Delta^{-1}[\rho] \right\} + \mathcal{O}(\hbar^2), \tag{4.42}$$

$$\boxed{U(\rho) = V(\rho) - \frac{i\hbar}{2\Omega} \ln\det\left\{ i\Delta^{-1}[\rho] \right\} + \mathcal{O}(\hbar^2).} \tag{4.43}$$

The determinant is evaluated using $\ln\det = \mathrm{Tr}\ln$,

$$\begin{aligned} \ln\det\left\{ i\Delta^{-1}[\rho] \right\} &= \mathrm{Tr}\ln\left\{ i\Delta^{-1}[\rho] \right\} \\ &= \int \frac{d^4k}{(2\pi)^4} \ln\langle k|\, i\Delta^{-1}[\rho]\, |k\rangle, \end{aligned} \tag{4.44}$$

so that

$$U(\rho) = V(\rho) - \frac{i\hbar}{2} \int \frac{d^4k}{(2\pi)^4}\, \Omega^{-1} \ln\langle k|\, i\Delta^{-1}[\rho]\, |k\rangle + \mathcal{O}(\hbar^2). \tag{4.45}$$

To evaluate $\Omega^{-1}\langle k|\, i\Delta^{-1}[\rho]\, |k\rangle$, expand $\phi(x) = \rho + \phi_1(x)$ with ρ constant and $\partial V/\partial\phi|_{\phi=\rho} = 0$,

$$\begin{aligned} S[\phi] &= \int d^4x \left[\frac{1}{2}\, \partial_\mu\phi_1\, \partial^\mu\phi_1 - V(\phi_1 + \rho) \right] \\ &= -\frac{1}{2} \int d^4x\; \phi_1 \left[\Box + V''(\rho) \right] \phi_1 + \cdots . \end{aligned} \tag{4.46}$$

By (4.31),

$$\begin{aligned} \langle x|\, i\Delta^{-1}[\rho]\, |y\rangle &= \left. \frac{\delta^2 S[\phi]}{\delta\phi_1(x)\, \delta\phi_1(y)} \right|_{\phi_1=0} \\ &= -\left[\Box + V''(\rho) \right] \delta^4(x - y). \end{aligned} \tag{4.47}$$

Taking the matrix element between momentum eigenstates (see Appendix E),

$$\Omega^{-1}\langle k|\, i\Delta^{-1}[\rho]\,|k\rangle = k^2 - V''(\rho). \tag{4.48}$$

Substituting into (4.45) and rotating to Euclidean momentum space via $k^0 \to ik_E^0$, $k^2 \to -k_E^2$, $d^4k = i\, d^4k_E$,

$$U(\rho) = V(\rho) + \frac{\hbar}{2}\int \frac{d^4k_E}{(2\pi)^4} \ln\left[k_E^2 + V''(\rho)\right] + \mathcal{O}(\hbar^2), \tag{4.49}$$

where the $i\pi$ contribution from $\ln(-k_E^2 - V'')$ is a field-independent constant and has been dropped. The momentum integral is evaluated in Appendix F, with the result

$$\frac{\hbar}{2}\int \frac{d^4k_E}{(2\pi)^4} \ln(k_E^2 + V'') \;=\; \hbar\left[\frac{\Lambda^2}{32\pi^2}V''(\rho) + \frac{[V''(\rho)]^2}{64\pi^2}\left(-\frac{1}{2} + \ln\frac{V''(\rho)}{\Lambda^2}\right) + \mathcal{O}\left(\frac{1}{\Lambda^2}\right)\right]. \tag{4.50}$$

We can therefore write

$$U(\rho) = V(\rho) + \hbar\, V_1(\rho) + \mathcal{O}(\hbar^2), \tag{4.51}$$

where the one-loop correction is

$$V_1(\rho) = \frac{\Lambda^2}{32\pi^2}V''(\rho) + \frac{[V''(\rho)]^2}{64\pi^2}\left[-\frac{1}{2} + \ln\frac{V''(\rho)}{\Lambda^2}\right]. \tag{4.52}$$

To separate the divergent from the convergent part, introduce an arbitrary renormalization scale μ via

$$\ln\frac{V''}{\Lambda^2} = \ln\frac{V''}{\mu^2} + \ln\frac{\mu^2}{\Lambda^2}, \tag{4.53}$$

so that

$$\begin{aligned} V_1(\rho) &= \frac{\Lambda^2}{32\pi^2}V''(\rho) + \frac{(V''(\rho))^2}{64\pi^2}\left[-\frac{1}{2} + \ln\frac{V''}{\mu^2} + \ln\frac{\mu^2}{\Lambda^2}\right] \\ &= \underbrace{\frac{1}{32\pi^2}\left[\Lambda^2 V''(\rho) - \frac{1}{2}[V''(\rho)]^2 \ln\frac{\Lambda^2}{\mu^2}\right]}_{\text{divergent}} + \underbrace{\frac{(V'')^2}{64\pi^2}\left[-\frac{1}{2} + \ln\frac{V''}{\mu^2}\right]}_{\text{convergent}}. \end{aligned} \tag{4.54}$$

The divergent part must be removed by renormalization, which we turn to next.

4.6 Renormalization

The theory is renormalizable if the counterterms needed to cancel the divergences are of the same form as terms already present in the Lagrangian. For a classical potential $V(\rho)$ that is a polynomial of degree n, the second derivative $V''(\rho)$ is of degree $n-2$, and the one-loop correction $V_1(\rho)$ in (4.52) is of degree $2(n-2) = 2n-4$ from the $(V'')^2$ term. Renormalizability then requires

$$2n - 4 \le n \quad\Rightarrow\quad n \le 4, \tag{4.55}$$

a condition satisfied by our choice of $V(\rho)$.

The bare parameters are split into renormalized parameters and counterterms,

$$\alpha_0 = \alpha_1 + \delta\alpha, \tag{4.56}$$
$$m_0^2 = m_1^2 + \delta m^2, \tag{4.57}$$

where $\delta\alpha$ and δm^2 are divergent in perturbation theory but α_1 and m_1^2 are required to be finite. Since the counterterms only arise at the quantum level, $\delta\alpha$, $\delta m^2 \sim \mathcal{O}(\hbar)$. The Lagrangian becomes

$$\mathcal{L}(x) = \frac{1}{2}\,\partial_\mu\phi\,\partial^\mu\phi - V(\phi) + \underbrace{\frac{\delta m^2}{2}\phi^2 - \frac{\delta\alpha}{4!}\phi^4}_{\text{counterterms}}, \tag{4.58}$$

with the renormalized classical potential

$$V(\phi) = \frac{\alpha_1}{4!}\phi^4 - \frac{m_1^2}{4}\phi^2. \tag{4.59}$$

The classical vacuum value ρ_0 is found from

$$\frac{dV(\phi)}{d\phi} = \frac{\alpha_1}{3!}\phi^3 - \frac{m_1^2}{2}\phi = 0$$
$$\Rightarrow \phi\left(\frac{\alpha_1}{3}\phi^2 - m_1^2\right) = 0, \tag{4.60}$$

giving $\phi = 0$ or

$$\rho_0 = m_1\left(\frac{3}{\alpha_1}\right)^{\frac{1}{2}}. \tag{4.61}$$

Since the counterterms are of order $\hbar$, they contribute at the same order as the one-loop correction and the full effective potential is

$$U(\rho) = V(\rho) + \hbar\,V_1(\rho) + \frac{\delta\alpha}{4!}\rho^4 - \frac{\delta m^2}{2}\rho^2. \tag{4.62}$$

Choosing $\delta\alpha$ and δm^2 to precisely cancel the divergent part of $V_1(\rho)$,

$$\delta\alpha = \frac{3\alpha_1^2\hbar}{32\pi^2}\ln\frac{\Lambda^2}{\mu^2}, \tag{4.63}$$
$$\delta m^2 = \frac{\alpha_1\hbar}{32\pi^2}\left(\Lambda^2 + \frac{m_1^2}{2}\ln\frac{\Lambda^2}{\mu^2}\right), \tag{4.64}$$

where the right-hand sides are ambiguous up to finite additive terms, the effective potential becomes

$$U(\rho) = V(\rho) + \frac{\hbar}{64\pi^2}\left[V''(\rho)\right]^2\left[-\frac{1}{2} + \ln\frac{V''(\rho)}{\mu^2}\right]. \tag{4.65}$$

4.7 Physical Parameters from the Effective Potential

We now extract the VEV, the renormalized mass, and the renormalized coupling constant from (4.65). Setting $k \equiv \hbar/(32\pi^2)$ for brevity, the successive derivatives of $U(\rho)$ are computed using $V = \frac{\alpha_1}{4!}\rho^4 - \frac{m_1^2}{4}\rho^2$, so $V' = \frac{\alpha_1}{3!}\rho^3 - \frac{m_1^2}{2}\rho$, $V'' = \frac{\alpha_1}{2}\rho^2 - \frac{m_1^2}{2}$, $V''' = \alpha_1\rho$, $V'''' = \alpha_1$, $V''''' = 0$.

Differentiating (4.65),

$$U' = V' + \frac{\hbar}{64\pi^2}\left[2V''V'''\left(-\frac{1}{2} + \ln\frac{V''}{\mu^2}\right) + (V'')^2 \cdot \frac{1}{V''} \cdot \frac{V'''}{\mu^2} \cdot \mu^2\right]$$
$$= V' + \frac{\hbar}{64\pi^2}\left[-V''V''' + 2V''V''' \ln\frac{V''}{\mu^2} + V''V'''\right], \quad (4.66)$$

$$\Rightarrow U' = V' + k\, V''V''' \ln\frac{V''}{\mu^2}. \quad (4.67)$$

Differentiating again,

$$U'' = V'' + k\left[(V''')^2 \ln\frac{V''}{\mu^2} + V''V'''' \ln\frac{V''}{\mu^2} + V''V''' \cdot \frac{1}{V''} \cdot \frac{V'''}{\mu^2} \cdot \mu^2\right], \quad (4.68)$$

$$\Rightarrow U'' = V'' + k\left[\left\{(V''')^2 + \alpha_1 V''\right\} \ln\frac{V''}{\mu^2} + (V''')^2\right]. \quad (4.69)$$

The third derivative,

$$U''' = V''' + k\Bigg[\left\{(V''')^2 + \alpha_1 V''\right\}' \ln\frac{V''}{\mu^2}$$
$$+ \left\{(V''')^2 + \alpha_1 V''\right\}\left(\ln\frac{V''}{\mu^2}\right)' + 2V'''V''''\Bigg]$$
$$= V''' + k\left[(2V'''V'''' + \alpha_1 V''') \ln\frac{V''}{\mu^2} + \left\{(V''')^2 + \alpha_1 V''\right\}\frac{V'''}{V''} + 2\alpha_1 V'''\right], \quad (4.70)$$

$$\Rightarrow U''' = V''' + k\left[3\alpha_1 V''' \ln\frac{V''}{\mu^2} + \frac{(V''')^3}{V''} + 3\alpha_1 V'''\right]. \quad (4.71)$$

The fourth derivative,

$$U'''' = V'''' + k\Bigg[3\alpha_1 V'''' \ln\frac{V''}{\mu^2} + 3\alpha_1 V''' \cdot \frac{1}{V''} \cdot \frac{V'''}{\mu^2} \cdot \mu^2$$
$$+ \frac{3(V''')^2 V''''}{V''} - \frac{(V''')^3}{(V'')^2} \cdot V''' + 3\alpha_1 V''''\Bigg]$$
$$= \alpha_1 + k\left[3\alpha_1^2 \ln\frac{V''}{\mu^2} + 3\alpha_1\frac{(V''')^2}{V''} + \frac{3\alpha_1 (V''')^2}{V''} - \frac{(V''')^4}{(V'')^2} + 3\alpha_1^2\right]$$
$$= \alpha_1 + k\left[3\alpha_1^2 \ln\frac{V''}{\mu^2} + \frac{6\alpha_1 (V''')^2}{V''} - \frac{(V''')^4}{(V'')^2} + 3\alpha_1^2\right], \quad (4.72)$$

$$\Rightarrow U'''' = \alpha_1 + k\alpha_1^2\left[3\ln\frac{V''}{\mu^2} + \frac{6\alpha_1\rho^2}{V''} - \frac{\alpha_1^2\rho^4}{(V'')^2} + 3\right]. \quad (4.73)$$

The VEV

The minimum $\langle\phi\rangle = \rho_0 + \rho_1$ is found by expanding $U'(\rho_0 + \rho_1) = 0$ to first order,

$$U'(\rho_0) + \rho_1 U''(\rho_0) = 0 \quad \Rightarrow \quad \rho_1 = -\frac{U'(\rho_0)}{U''(\rho_0)}. \quad (4.74)$$

We first evaluate $V''(\rho_0)$ and $V'''(\rho_0)$ at the classical minimum $\rho_0^2 = 3m_1^2/\alpha_1$,

$$V''(\rho_0) = \frac{\alpha_1}{2}\rho_0^2 - \frac{m_1^2}{2} = \frac{\alpha_1}{2}\cdot\frac{3m_1^2}{\alpha_1} - \frac{m_1^2}{2} = \frac{3m_1^2}{2} - \frac{m_1^2}{2} = m_1^2, \tag{4.75}$$

$$\Rightarrow V''(\rho_0) = m_1^2. \tag{4.76}$$

Also, $V'(\rho_0) = 0$ by definition of ρ_0, so from (4.67),

$$U'(\rho_0) = k\, m_1^2\, \alpha_1\, \rho_0 \ln\frac{m_1^2}{\mu^2}. \tag{4.77}$$

For $U''(\rho_0)$, the $\mathcal{O}(\alpha_1^2)$ terms are dropped,

$$\begin{aligned} U''(\rho_0) &= m_1^2 + k\left[(\alpha_1^2\rho_0^2 + \alpha_1 m_1^2)\ln\frac{m_1^2}{\mu^2} + \alpha_1^2\rho_0^2\right] \\ &= m_1^2 + k\,\alpha_1 m_1^2 \ln\frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2), \end{aligned} \tag{4.78}$$

$$\Rightarrow U''(\rho_0) = m_1^2\left(1 + \alpha_1 k \ln\frac{m_1^2}{\mu^2}\right) + \mathcal{O}(\alpha_1^2). \tag{4.79}$$

Hence,

$$\begin{aligned} \rho_1 &= -\frac{k\, m_1^2\, \alpha_1\, \rho_0 \ln\frac{m_1^2}{\mu^2}}{m_1^2\left(1 + \alpha_1 k \ln\frac{m_1^2}{\mu^2}\right)} \\ &= -k\,\alpha_1\,\rho_0 \ln\frac{m_1^2}{\mu^2}\left(1 + \alpha_1 k \ln\frac{m_1^2}{\mu^2}\right)^{-1} \\ &= -k\alpha_1\rho_0 \ln\frac{m_1^2}{\mu^2}\left[1 - \alpha_1 k \ln\frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2)\right] \\ &= -k\alpha_1\rho_0 \ln\frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2), \end{aligned} \tag{4.80}$$

$$\frac{\rho_1}{\rho_0} = -\frac{\hbar\alpha_1}{32\pi^2}\ln\frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2). \tag{4.81}$$

Setting $\hbar = 1$, the full VEV is

$$\boxed{\langle\phi\rangle = m_1\left(\frac{3}{\alpha_1}\right)^{\frac{1}{2}}\left[1 - \frac{\alpha_1}{32\pi^2}\ln\frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2)\right].} \tag{4.82}$$

The Renormalized Mass

The renormalized mass is $m^2 \equiv U''(\rho_0 + \rho_1)$. Expanding,

$$m^2 = U''(\rho_0) + \rho_1\, U'''(\rho_0). \tag{I}$$

From (4.71), dropping $\mathcal{O}(\alpha_1^2)$ terms,

$$\begin{aligned} U'''(\rho_0) &= \alpha_1\rho_0 + k\left[3\alpha_1^2\rho_0 \ln\frac{m_1^2}{\mu^2} + \frac{(\alpha_1\rho_0)^3}{m_1^2} + 3\alpha_1^2\rho_0\right] \\ &= \alpha_1\rho_0 + \mathcal{O}(\alpha_1^2), \end{aligned} \tag{II}$$

$$U'''(\rho_0) = \alpha_1\rho_0 + k\left[3\alpha_1^2\rho_0 \ln\frac{m_1^2}{\mu^2}\right]. \tag{III}$$

Also,

$$U''(\rho_0) = m_1^2 + k\left[(\alpha_1^2\rho_0^2 + \alpha_1 m_1^2)\ln\frac{m_1^2}{\mu^2} + \alpha_1^2\rho_0^2\right]. \tag{4.83}$$

Substituting (II) and (III) into (I), using $\rho_1 = -k\alpha_1\rho_0 \ln\frac{m_1^2}{\mu^2}$ and $\rho_1 U'''(\rho_0) = -k\alpha_1\rho_0 \ln\frac{m_1^2}{\mu^2}\cdot 3m_1^2 + \mathcal{O}(\alpha_1^2)$,

$$\begin{aligned}
m^2 &= m_1^2\left(1 + 4\alpha_1 k\ln\frac{m_1^2}{\mu^2} + 3\alpha_1 k\right) - 3m_1^2\alpha_1 k\ln\frac{m_1^2}{\mu^2} + \mathcal{O}(\alpha_1^2) \\
&= m_1^2\left[1 + 4\alpha_1 k\ln\frac{m_1^2}{\mu^2} + 3\alpha_1 k - 3\alpha_1 k\ln\frac{m_1^2}{\mu^2}\right] + \mathcal{O}(\alpha_1^2) \\
&= m_1^2\left[1 + \alpha_1 k\ln\frac{m_1^2}{\mu^2} + 3\alpha_1 k\right] + \mathcal{O}(\alpha_1^2) \\
&= m_1^2\left[1 + 3\alpha_1 k\left(1 + \frac{1}{3}\ln\frac{m_1^2}{\mu^2}\right)\right] + \mathcal{O}(\alpha_1^2),
\end{aligned} \tag{4.84}$$

$$\Rightarrow \quad \boxed{m^2 = m_1^2\left[1 + \frac{3\alpha_1}{32\pi^2}\left(1 + \frac{1}{3}\ln\frac{m_1^2}{\mu^2}\right) + \mathcal{O}(\alpha_1^2)\right].} \tag{4.85}$$

The Renormalized Coupling Constant

The renormalized coupling is $\alpha \equiv U''''(\rho_0 + \rho_1)$. Evaluating (4.73) at $\rho = \rho_0$ and keeping terms to $\mathcal{O}(\alpha_1)$,

$$\begin{aligned}
\alpha &= \alpha_1 + k\alpha_1^2\left[3\ln\frac{V''}{\mu^2} + \frac{6\alpha_1\rho^2}{V''} - \frac{\alpha_1^2\rho^4}{(V'')^2} + 3\right] \\
&= \alpha_1\left[1 + 3\alpha_1 k\left(1 + \ln\frac{V''}{\mu^2}\right) + \mathcal{O}(\alpha_1^2)\right],
\end{aligned} \tag{4.86}$$

$$\Rightarrow \quad \boxed{\alpha = \alpha_1\left[1 + \frac{3\alpha_1}{32\pi^2}\left(1 + \ln\frac{m_1^2}{\mu^2}\right) + \mathcal{O}(\alpha_1^2)\right].} \tag{4.87}$$

4.8 The Massless Case and the Running Coupling Constant

We now consider the massless limit $m_1 = 0$, in which

$$V' = \frac{\alpha_1}{6}\rho^3, \qquad\qquad V'' = \frac{\alpha_1\rho^2}{2}. \tag{4.88}$$

From (4.73),

$$\begin{aligned}
U''''(\rho) &= \alpha_1 + k\alpha_1^2\left[3\ln\frac{\alpha_1\rho^2}{2\mu^2} + \frac{6\alpha_1\rho^2}{\frac{\alpha_1\rho^2}{2}} - \frac{\alpha_1^2\rho^4}{\frac{\alpha_1^2\rho^4}{4}} + 3\right] \\
&= \alpha_1 + k\alpha_1^2\left[3\ln\frac{\alpha_1\rho^2}{2\mu^2} + 12 - 4 + 3\right],
\end{aligned} \tag{4.89}$$

$$\Rightarrow U''''(\rho) = \alpha_1 + k\alpha_1^2 \left[11 + 3\ln\frac{\alpha_1\rho^2}{2\mu^2}\right]. \tag{4.90}$$

Setting $\alpha_1 \equiv \alpha$ and defining the running coupling constant $\alpha(\rho) \equiv U''''(\rho)$,

$$\alpha(\rho) = \alpha + \frac{k\alpha^2}{1}\left[11 + 3\ln\frac{\alpha\rho^2}{2\mu^2}\right]. \tag{4.91}$$

We introduce a new scale μ_0 defined by the renormalization condition $U''''(\mu_0) = \alpha$, which gives

$$\begin{aligned} \alpha &= \alpha + k\alpha^2\left[11 + 3\ln\frac{\alpha\mu_0^2}{2\mu^2}\right] \\ \Rightarrow \quad \ln\frac{\alpha\mu_0^2}{2\mu^2} &= -\frac{11}{3}. \end{aligned} \tag{4.92}$$

Using this to eliminate the explicit $\ln(\alpha/2\mu^2)$ dependence,

$$\begin{aligned} \alpha(\rho) &= \alpha + k\alpha^2\left[3\cdot\frac{11}{3} + 3\ln\frac{\alpha\rho^2}{2\mu^2}\right] \\ &= \alpha + k\alpha^2\left[-3\ln\frac{\alpha\mu_0^2}{2\mu^2} + 3\ln\frac{\alpha\rho^2}{2\mu^2}\right] \\ &= \alpha + 3k\alpha^2\ln\left(\frac{\frac{\alpha\rho^2}{2\mu^2}}{\frac{\alpha\mu_0^2}{2\mu^2}}\right), \end{aligned} \tag{4.93}$$

$$\Rightarrow \quad \boxed{\alpha(\rho) = \alpha + \frac{3k\alpha^2}{32\pi^2}\ln\frac{\rho^2}{\mu_0^2},} \tag{4.94}$$

where $\alpha = \alpha(\mu_0)$ is the value of the running coupling at $\rho = \mu_0$. Inverting,

$$\frac{1}{\alpha(\rho)} = \frac{1}{\alpha(\mu_0) + \frac{3k\alpha^2(\mu_0)}{32\pi^2}\ln\frac{\rho^2}{\mu_0^2}} = \frac{1}{\alpha(\mu_0)}\left[1 + \frac{3k\alpha(\mu_0)}{32\pi^2}\ln\frac{\rho^2}{\mu_0^2}\right]^{-1}, \tag{4.95}$$

$$\Rightarrow \quad \frac{1}{\alpha(\rho)} = \frac{1}{\alpha(\mu_0)}\left(1 - \frac{3k\alpha(\mu_0)}{32\pi^2}\ln\frac{\rho^2}{\mu_0^2}\right) + \text{higher order.} \tag{4.96}$$

4.9 Spontaneous Symmetry Breaking by Radiative Corrections?

A natural question arises: can SSB occur in the massless theory purely due to radiative corrections, generating a nonzero $\langle\phi\rangle$ dynamically? From (4.67) with $m_1 = 0$,

$$\begin{aligned} U'(\rho) &= V' + k\,V''V'''\ln\frac{V''}{\mu^2} \\ &= \frac{\alpha}{6}\rho^3 + k\,\frac{\alpha\rho^2}{2}\cdot\alpha\rho\cdot\ln\frac{\alpha\rho^2}{2\mu^2} \\ &= \frac{\rho^3}{6}\left[\alpha + 3k\alpha^2\ln\frac{\alpha\rho^2}{2\mu^2}\right]. \end{aligned} \tag{4.97}$$

Writing $\ln \frac{\alpha\rho^2}{2\mu^2} = \ln \frac{\rho^2}{\mu^2} + \ln \frac{\alpha}{2}$, where the last term is a field-independent constant,

$$U'(\rho) = \frac{\rho^3}{6}\left[\alpha + \frac{3\hbar\alpha^2}{32\pi^2}\left(\ln \frac{\rho^2}{\mu^2} + \text{const}\right)\right]. \tag{4.98}$$

Setting $U'(\langle\phi\rangle) = 0$ for a nontrivial root gives

$$\begin{aligned} & \alpha + \frac{3k\alpha^2}{32\pi^2}\ln\frac{\langle\phi\rangle^2}{\mu^2} = 0 \\ \Rightarrow\quad & \alpha\left(1 + \frac{3k\alpha}{32\pi^2}\ln\frac{\langle\phi\rangle^2}{\mu^2}\right) = 0 \\ \Rightarrow\quad & \hbar\alpha\ln\frac{\langle\phi\rangle^2}{\mu^2} = -\frac{32\pi^2}{3} + \mathcal{O}(\hbar). \end{aligned} \tag{4.99}$$

This root must be rejected. The loop expansion is valid only when $\hbar\alpha$ is small, but (4.99) requires

$$\hbar\alpha\ln\frac{\langle\phi\rangle^2}{\mu^2} \sim \mathcal{O}(1), \tag{4.100}$$

meaning higher-order terms are not small and cannot be neglected. The minimum lies outside the domain of validity of the one-loop approximation. Let us explain this in slightly more details. The loop expansion is an expansion in powers of $\hbar$, with the L-loop contribution carrying a factor of $\hbar^L$. The one-loop approximation is therefore valid when the one-loop correction is small compared to the tree-level term,

$$\frac{\hbar V_1(\rho)}{V(\rho)} \ll 1, \tag{4.101}$$

or more precisely in this case, when

$$\hbar\alpha\ln\frac{\langle\phi\rangle^2}{\mu^2} \ll 1. \tag{4.102}$$

The spurious root requires precisely the opposite, $\hbar\alpha\ln(\langle\phi\rangle^2/\mu^2) \sim \mathcal{O}(1)$, so that the logarithm is of order $1/(\hbar\alpha)$ and $\langle\phi\rangle$ sits exponentially far from μ. At such field values the two-loop contribution goes as

$$(\hbar\alpha)^2 \cdot \left(\ln\frac{\langle\phi\rangle^2}{\mu^2}\right)^2 \sim \mathcal{O}(1), \tag{4.103}$$

and similarly for all higher loops, so every order contributes equally and truncating at one loop is unjustified. The root is therefore rejected, not

of validity of the approximation.

The technical reason for this failure is that the two terms in (4.98) are of order α and α^2 respectively, and cannot cancel each other within the range of validity of perturbation theory: an $\mathcal{O}(\alpha)$ quantity cannot cancel an $\mathcal{O}(\alpha^2)$ quantity. However, if the model were modified so that α and α^2 were replaced by two independent coupling constants, the cancellation could occur and one would obtain a genuine theory of SSB by radiative corrections. The simplest such model is massless scalar electrodynamics, to which we turn in the next section.

Chapter 5

Massless Scalar Electrodynamics and Dimensional Transmutation

In renormalized massless scalar electrodynamics, the scalar field acquires an arbitrary vacuum expectation value $\langle\phi\rangle$ even though classically it vanishes. This means $\langle\phi\rangle$ emerges as a spontaneously generated physical mass scale, with both scalar and vector particles developing dynamical masses proportional to it. This phenomenon is called *dimensional transmutation*: a theory with no classical mass scale develops a preferred mass scale through radiative corrections.

5.1 The Lagrangian and Unitary Gauge

The Lagrangian of massless scalar electrodynamics is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + [(\partial_\mu - ieA_\mu)\phi^*]\,[(\partial^\mu + ieA^\mu)\phi] - \frac{\lambda_0}{6}(\phi^*\phi)^2, \tag{5.1}$$

where e is the renormalized coupling constant and λ_0 is the bare scalar coupling, necessary to renormalize the scalar-scalar scattering amplitude. Transforming away the phase of ϕ by going to unitary gauge and rescaling $\phi \to \phi/\sqrt{2}$, the Lagrangian becomes

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}e^2 A_\mu A^\mu \phi^2 + \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{\lambda_0}{4!}\phi^4, \tag{5.2}$$

subject to the Lorenz condition $\partial_\mu A^\mu = 0$. The corresponding action is

$$S[A,\phi] = \int d^4x \left[\frac{1}{2}A^\mu(\partial^2 + e^2\phi^2)A_\mu - \frac{1}{2}\phi\,\partial^2\phi - \frac{\lambda_0}{4!}\phi^4\right]. \tag{5.3}$$

5.2 Integrating Out the Vector Field

To obtain an effective action for ϕ alone, we integrate out the vector field,

$$\exp iS[\phi] = \mathcal{N}\int (\mathcal{D}A)\,\delta[\partial_\mu A^\mu]\,\exp iS[A,\phi], \tag{5.4}$$

where $\delta[\partial_\mu A^\mu]$ enforces the Lorenz constraint. Each transverse component of A_μ contributes a Gaussian determinant,

$$\begin{aligned}\left[\det(\partial^2 + e^2\phi^2)\right]^{-\frac{1}{2}} &= \exp\left[-\frac{1}{2}\ln\det(\partial^2 + e^2\phi^2)\right] \\ &= \exp\left[-\frac{1}{2}\mathrm{Tr}\ln(\partial^2 + e^2\phi^2)\right]. \end{aligned} \tag{5.5}$$

There are three such transverse components, so the effective action becomes

$$S[\phi] = -\int d^4x \left(\frac{1}{2}\phi\,\partial^2\phi + \frac{\lambda_0}{4!}\phi^4\right) - \frac{3}{2}\mathrm{Tr}\ln(\partial^2 + e^2\phi^2). \tag{5.6}$$

The one-loop effective potential for ϕ, to lowest order in λ_0, is then

$$U(\rho) = \frac{\lambda_0}{4!}\rho^4 + \frac{3}{2}\int \frac{d^4k_E}{(2\pi)^4}\ln(k_E^2 + e^2\rho^2), \tag{5.7}$$

where the second term sums all one-loop graphs with external ϕ lines and vector boson running in the loop, since we integrated out A_μ. Scalar loops contribute only at higher order.

5.3 The Renormalized Effective Potential

Using the machinery developed in the previous chapter, specifically Eq.(4.65), the renormalized effective potential in this case takes the form

$$U(\rho) = \rho^4\left[\frac{\lambda}{4!} + \frac{3e^4}{64\pi^2}\left(-\frac{1}{2} + \ln\frac{e^2\rho^2}{\mu^2}\right)\right], \tag{5.8}$$

where λ is the running coupling constant at the renormalization point μ. Note that e^4 is a one-loop result and λ is a tree-level result, but they can be comparable since they are independent coupling constants. This is what allows the cancellation that was impossible in the pure scalar case.

5.4 Spontaneous Symmetry Breaking and Particle Masses

Differentiating (5.8),

$$\begin{aligned} U'(\rho) &= 4\rho^3\left[\cdots\right] + \rho^4\cdot\frac{3e^4}{64\pi^2}\cdot\frac{1}{\frac{e^2\rho^2}{\mu^2}}\cdot\frac{2\rho e^2}{\mu^2} \\ &= 4\rho^3\left[\cdots\right] + \frac{3e^4}{64\pi^2}\rho^3\cdot 2 \\ &= \rho^3\left[4\cdot\frac{\lambda}{4!} + 4\cdot\frac{3e^4}{64\pi^2}\left(-\frac{1}{2}\right) + 4\cdot\frac{3e^4}{64\pi^2}\ln\frac{e^2\rho^2}{\mu^2} + \frac{3e^4}{64\pi^2}\cdot 2\right], \end{aligned} \tag{5.9}$$

$$\Rightarrow U'(\rho) = 4\rho^3\left[\frac{\lambda}{4!} + \frac{3e^4}{64\pi^2}\ln\frac{e^2\rho^2}{\mu^2}\right]. \tag{5.10}$$

Setting $U'(\langle\phi\rangle) = 0$ for a nontrivial root,

$$\begin{aligned} \frac{\lambda}{4!} + \frac{3e^4}{64\pi^2} \ln \frac{e^2\langle\phi\rangle^2}{\mu^2} &= 0 \\ \Rightarrow \quad \frac{3e^4}{64\pi^2} \ln \frac{e^2\langle\phi\rangle^2}{\mu^2} &= -\frac{\lambda}{4!}. \end{aligned} \tag{5.11}$$

Unlike the pure scalar case, this root is within the domain of validity of the approximation since λ and e^4 are independent and can be chosen to make the two terms comparable without requiring the logarithm to be large. Using this condition to eliminate μ from (5.8),

$$\begin{aligned} U(\rho) &= \rho^4 \left[-\frac{3e^4}{64\pi^2} \ln \frac{e^2\langle\phi\rangle^2}{\mu^2} + \frac{3e^4}{64\pi^2}\left(-\frac{1}{2}\right) + \frac{3e^4}{64\pi^2} \ln \frac{e^2\rho^2}{\mu^2} \right] \\ &= \frac{3e^4\rho^4}{64\pi^2} \left[-\frac{1}{2} + \ln \frac{\rho^2}{\langle\phi\rangle^2} \right]. \end{aligned} \tag{5.12}$$

The scalar particle mass follows from $m_S^2 \equiv U''(\langle\phi\rangle)$. Setting $a \equiv 3e^4/(64\pi^2)$ for brevity, we differentiate $U(\rho) = a\rho^4 \left[-\frac{1}{2} + \ln \frac{\rho^2}{\langle\phi\rangle^2} \right]$,

$$\begin{aligned} U'(\rho) &= a \cdot 4\rho^3 \left[-\frac{1}{2} + \ln \frac{\rho^2}{\langle\phi\rangle^2} \right] + a\rho^4 \cdot \frac{\langle\phi\rangle^2}{\rho^2} \cdot \frac{2\rho}{\langle\phi\rangle^2} \\ &= 2a\rho^3 \left[2\left(-\frac{1}{2} + \ln \frac{\rho^2}{\langle\phi\rangle^2} \right) + 1 \right] \\ &= 4a\rho^3 \ln \frac{\rho^2}{\langle\phi\rangle^2}. \end{aligned} \tag{5.13}$$

Differentiating once more,

$$\begin{aligned} U''(\rho) &= 12a\rho^2 \ln \frac{\rho^2}{\langle\phi\rangle^2} + 4a\rho^3 \cdot \frac{\langle\phi\rangle^2}{\rho^2} \cdot \frac{2\rho}{\langle\phi\rangle^2} \\ &= 12a\rho^2 \ln \frac{\rho^2}{\langle\phi\rangle^2} + 8a\rho^2. \end{aligned} \tag{5.14}$$

Evaluating at $\rho = \langle\phi\rangle$,

$$\begin{aligned} U''(\langle\phi\rangle) &= 12a\langle\phi\rangle^2 \ln \frac{\langle\phi\rangle^2}{\langle\phi\rangle^2} + 8a\langle\phi\rangle^2 \\ &= 8 \cdot \frac{3e^4}{64\pi^2}\langle\phi\rangle^2, \end{aligned} \tag{5.15}$$

$$\Rightarrow \quad \boxed{m_S^2 = \frac{3e^4}{8\pi^2}\langle\phi\rangle^2.} \tag{5.16}$$

The vector field acquires mass through the Higgs mechanism,

$$m_V^2 = e^2\langle\phi\rangle^2. \tag{5.17}$$

Both masses are proportional to $\langle\phi\rangle$, confirming that the dynamically generated VEV sets the mass scale of the theory.

5.5 Dimensional Transmutation

To renormalize the theory we introduce a momentum cutoff, which represents a hidden scale. After renormalization this hidden scale is replaced by an arbitrary finite renormalization point μ. In a classically massless theory one might expect all choices of μ to be equivalent, but what we have found is that there is a preferred renormalization point: the one at which the particles develop mass dynamically. Dimensional transmutation refers precisely to this emergence of a preferred scale in a theory with no intrinsic mass, whereby the infinite cutoff momentum becomes transmuted into a finite physical mass. This mechanism is widely believed to operate in QCD with massless quarks, where no intrinsic mass scale exists in the Lagrangian yet hadrons acquire mass dynamically.

Chapter 6

One-Loop Effective Potential in Non-Abelian Gauge Theories

The mechanism of spontaneous symmetry breaking via radiative corrections, which we saw explicitly in massless scalar electrodynamics, works equally well for non-Abelian gauge theories. The computation of the one-loop effective potential is conceptually the same as in the scalar case, though the technical details are more involved due to the richer field content. We follow the treatment of Brandenberger closely.

6.1 The Theory

Consider a non-Abelian gauge theory with gauge group G and generators τ_a, a set of Dirac bispinor fields ψ^a, and a set of scalar fields φ_i, each forming a representation space for G. The Lagrangian is

$$\mathcal{L}(A,\psi,\varphi) = -\frac{1}{4}\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu}) + i\,\mathrm{Tr}(\bar{\psi}\not{D}\psi) + \mathrm{Tr}(\bar{\psi}M\psi) - \mathrm{Tr}(\bar{\psi}\Gamma\varphi\psi) + \frac{1}{2}\mathrm{Tr}(D_\mu\varphi)^\dagger D^\mu\varphi - U(\varphi), \tag{6.1}$$

where the gauge field, field strength, and covariant derivative are

$$A_\mu = A_\mu^a\tau_a, \tag{6.2}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ig[A_\mu, A_\nu], \tag{6.3}$$

$$D_\mu = \partial_\mu + igA_\mu. \tag{6.4}$$

In D_μ, the τ_a are the matrices representing the generators in the appropriate representation space. The trace refers to the appropriate representation space: if the Higgs and Fermi fields are represented as column vectors, the trace reduces to the scalar product, e.g. $\mathrm{Tr}(D_\mu\varphi)^\dagger D^\mu\varphi = (D_\mu\varphi)^\dagger D^\mu\varphi$. The matrix Γ denotes the Yukawa coupling constants.

The Feynman rules in general involve extra fields, the ghost fields ω and $\bar{\omega}$, scalar fields obeying Fermi statistics (Faddeev and Popov, 1967). They arise as a consequence of the field-dependent functional determinant in the Faddeev-Popov ansatz for the generating functional. In covariant gauges, the ghost term in the effective Lagrangian is $\mathrm{Tr}\,\bar{\omega}\,\partial_\mu A^\mu\omega$, and contains no Higgs couplings to ghost fields. Hence to one-loop order no ghosts arise in our calculation of the effective potential. The one-loop effective potential therefore takes the form

$$V_{\text{eff}}(\bar{\varphi}) = U(\bar{\varphi}) + V^{(1)}_{\text{c.t.}} + V_{\text{1-loop}}, \tag{6.5}$$

where $V^{(1)}_{\text{c.t.}}$ contains all counterterms to order $\hbar$ and $V_{\text{1-loop}}$ is the sum of all one-loop graphs with external φ lines at zero momenta. By charge conservation, the particle type cannot change within a single loop, so the contributing one-loop diagrams are those with Higgs bosons, fermions, or gauge bosons running in the loop. The corresponding contributions are denoted $V_S^{(1)}$, $V_f^{(1)}$, and $V_g^{(1)}$.

6.2 Diagonalization of the Scalar Mass Matrix

The scalar sector of the theory is described by the Hessian of the potential, evaluated at the background field $\bar{\varphi}$,

$$W^{ij}(\bar{\varphi}) = \left.\frac{\partial^2 U(\varphi)}{\partial\varphi_i\,\partial\varphi_j}\right|_{\varphi=\bar{\varphi}}. \tag{6.6}$$

This matrix controls the quadratic fluctuations around the background. To see this explicitly, expand around the background $\varphi_i(x) = \bar{\varphi}_i + \eta_i(x)$, where η_i are small fluctuations. Since $\bar{\varphi}$ satisfies the classical equations of motion, the linear term vanishes and the quadratic Lagrangian is

$$\mathcal{L}_{\text{quad}} = \frac{1}{2}(\partial_\mu\eta_i)^2 - \frac{1}{2}\eta_i W^{ij}(\bar{\varphi})\eta_j. \tag{6.7}$$

If W^{ij} is not diagonal, the fields η_i are mixed and do not correspond to independent physical particles. To decouple them, we diagonalize W.

For an even quartic potential, the most general form is

$$U(\varphi) = \frac{1}{2}m_{ij}\varphi_i\varphi_j + \frac{1}{4!}\lambda_{ijkl}\varphi_i\varphi_j\varphi_k\varphi_l, \tag{6.8}$$

with $m_{ij} = m_{ji}$ and $\lambda_{ijkl} = \lambda_{(ijkl)}$. Computing the second derivative,

$$W^{ab}(\bar{\varphi}) = m_{ab} + \frac{1}{2}\lambda_{abkl}\bar{\varphi}_k\bar{\varphi}_l. \tag{6.9}$$

Since m_{ab} is symmetric and λ_{abkl} is symmetric in a, b, the matrix W^{ab} is real and symmetric. By the spectral theorem, any real symmetric matrix can be diagonalized by an orthogonal transformation: there exists an orthogonal matrix O such that

$$O^T W O = \text{diag}(w_1, \ldots, w_N). \tag{6.10}$$

Defining the rotated fields $\chi_i = O_{ij}\eta_j$, and using the fact that orthogonal transformations preserve the kinetic term, the quadratic Lagrangian becomes

$$\boxed{\mathcal{L}_{\text{quad}} = \sum_i \left[\frac{1}{2}(\partial_\mu\chi_i)^2 - \frac{1}{2}w_i\,\chi_i^2\right].} \tag{6.11}$$

Each mode χ_i is now an independent scalar with mass squared w_i, propagating freely without mixing. These are the physical Higgs modes.

As a concrete example, for the gauge-invariant quartic potential $U(\varphi) = \frac{1}{2}m^2\varphi_i\varphi_i + \frac{\lambda}{4!}(\varphi_i\varphi_i)^2$, choosing the background along a single direction $\bar{\varphi}_i = v\,\delta_{i1}$ gives

$$W^{11} = m^2 + \frac{\lambda}{2}v^2 \quad \text{(radial/Higgs mode)}, \tag{6.12}$$

$$W^{aa} = m^2 + \frac{\lambda}{6}v^2 \quad (N-1 \text{ degenerate transverse modes, } a \neq 1), \tag{6.13}$$

$$W^{ab} = 0 \quad \text{for } a \neq b, \tag{6.14}$$

so the matrix is already diagonal in this basis.

6.3 The Higgs Boson Contribution $V_S^{(1)}$

For $V_S^{(1)}$, the only change relative to the scalar field theory example is the presence of more than one scalar particle. The relevant diagrams are those of the single scalar case, now with one internal line associated with φ_i and the other with φ_j.

The effective coupling constant, which includes the $\bar{\varphi}$ factors for internal links and the Bose combinatorial factors, is

$$W^{ij} = \left.\frac{\partial^2 U(\varphi)}{\partial\varphi_i\,\partial\varphi_j}\right|_{\varphi=\bar{\varphi}}. \tag{6.15}$$

In the single scalar field theory, $W = \frac{1}{2}\lambda\bar{\varphi}^2$. If $U(\varphi)$ is an even quartic polynomial, $W(\bar{\varphi})$ is a quadratic form and can be diagonalized as shown above. In the diagonal basis, the different Higgs fields decouple and $V_S^{(1)}$ is obtained by adding the contributions of the new basis fields. Each individual term is given by the scalar Coleman-Weinberg integral. Neglecting the λ^2 corrections to terms already contained in $V_{\text{eff}}^{(0)}$, the cutoff-independent part of the one-loop integral for a single mode with $m^2 = \frac{\lambda}{2}\bar{\varphi}^2$ is

$$V_{\text{1-loop}}\big|_{\text{c.i.}} = \frac{1}{64\pi^2}\left(\frac{\lambda\bar{\varphi}^2}{2}\right)^2 \ln\frac{\lambda\bar{\varphi}^2}{2}, \tag{6.16}$$

where the divergent terms proportional to Λ^2 and $\ln\Lambda^2$ have been absorbed into renormalization of the mass and coupling constant respectively. Generalizing to the diagonal basis,

$$\begin{aligned} V_S^{(1)}(\bar{\varphi}) &= \frac{1}{64\pi^2}\sum_i (W^{ii})^2 \ln W^{ii} \\ &= \frac{1}{64\pi^2}\mathrm{Tr}(W^2 \ln W), \end{aligned} \tag{6.17}$$

which is basis-independent since the trace is invariant under orthogonal transformations.

6.4 The Fermion Contribution $V_f^{(1)}$

The fermion contribution $V_f^{(1)}$ is given by the diagrams in Fig. 3 of Brandenberger, for the case $M = 0$. Since the trace of an odd number of γ matrices vanishes, diagrams with an odd number of vertices are zero. With all indices inserted explicitly, the Yukawa coupling term from (6.1) is

$$\sum_{a,b,i} \bar{\varphi}_i\,\Gamma^i_{ab}\,\psi^a\psi^b. \tag{6.18}$$

The diagram with two external scalar field insertions $\bar{\varphi}_i$ and $\bar{\varphi}_j$ consists of a closed fermion loop with two internal massless propagators $\not{k}/k^2$ and two Yukawa vertices contributing Γ^i_{ab} and Γ^j_{ba}. Assembling all factors, the overall sign (-1) from the fermion loop and the symmetry factor $\frac{1}{2}$ combine to give

$$-\frac{1}{2}\int\frac{d^4k}{(2\pi)^4}\sum_{i,j,a,b}\mathrm{Tr}_S\,\bar{\varphi}_i\,\Gamma^i_{ab}\,\frac{\not{k}^2}{k^4}\,\Gamma^j_{ba}\,\bar{\varphi}_j. \tag{6.19}$$

Using $(\not{k})(\not{k}) = k^2$, the propagator product simplifies to $\not{k}^2/k^4 = 1/k^2$. The spinor trace then acts only on the identity in spinor space, giving $\mathrm{Tr}_S(\mathbf{1}) = 4$ in four dimensions. Defining the matrix $(\bar\varphi\,\Gamma)_{ab} \equiv \sum_i \bar\varphi_i\,\Gamma^i_{ab}$, the sum over indices becomes $\sum_{a,b}(\bar\varphi\,\Gamma)_{ab}(\bar\varphi\,\Gamma)_{ba} = \mathrm{Tr}_f(\bar\varphi\,\Gamma)^2$, and (6.19) reduces to

$$-\frac{1}{2}\int\frac{d^4k}{(2\pi)^4}\,\frac{4}{k^2}\,\mathrm{Tr}_f(\bar\varphi\,\Gamma)^2. \tag{6.20}$$

The graph with $2n$ vertices yields analogously

$$\frac{1}{2n}\int\frac{d^4k}{(2\pi)^4}\,\frac{1}{(k^2)^n}\,\mathrm{Tr}_f(\bar\varphi\,\Gamma)^{2n}, \tag{6.21}$$

so that summing over all diagrams,

$$V_f^{(1)}(\bar\varphi) = -4i\,\mathrm{Tr}_f\sum_{n=1}^{\infty}\int\frac{d^4k}{(2\pi)^4}\,\frac{1}{2n}\left[\frac{(\bar\varphi\,\Gamma)^2}{k^2}\right]^n. \tag{6.22}$$

This result is analogous to (6.16), and its evaluation follows the same procedure as in the scalar case. The cutoff-independent part gives

$$V_f^{(1)}(\bar\varphi) = -\frac{1}{64\pi^2}\,4\,\mathrm{Tr}_f\left[(\bar\varphi\,\Gamma)^4\ln(\bar\varphi\,\Gamma)^2\right]. \tag{6.23}$$

6.5 The Gauge Boson Contribution $V_g^{(1)}$

Finally we compute $V_g^{(1)}$, the gauge boson loop contribution. The only vertex contributing to one-loop order is the $A^2\varphi^2$ vertex, which stems from the following term in the Lagrangian density,

$$\frac{1}{2}\mathrm{Tr}(igA_\mu\varphi)^\dagger(igA^\mu\varphi) = \frac{1}{2}g_{\alpha\beta}\mathrm{Tr}\left[(\tau_\alpha A_\mu^\alpha\varphi)^\dagger(\tau_\beta A^{\beta\mu}\varphi)\right] = \frac{1}{2}A_\mu^\alpha M^2_{\alpha\beta}A^{\beta\mu}, \tag{6.24}$$

with the field-dependent gauge boson mass matrix

$$M^2_{\alpha\beta} = g_{\alpha\beta}\,\mathrm{Tr}\left[(\tau_\alpha\varphi)^\dagger\tau_\beta\varphi\right]. \tag{6.25}$$

A few comments on (6.25) are in order. First, if G is not a simple group, there will in general be a different coupling constant associated with each field A_μ^a. Second, the trace is in the Higgs representation space. Third, $M^2_{\alpha\beta}|_{\varphi=\bar\varphi}$ is the gauge boson mass matrix in the broken phase with vacuum expectation value $\bar\varphi$, and is a constant matrix in the sense of (6.6). Since in Landau gauge the gauge fields have three degrees of freedom (they have a Lie algebra trace), the calculation of $V_g^{(1)}$ is analogous to that of $V_S^{(1)}$, and

$$V_g^{(1)}(\bar\varphi) = \frac{3}{64\pi^2}\mathrm{Tr}(M^4\ln M^2)\big|_{\varphi=\bar\varphi}. \tag{6.26}$$

6.6 The Full One-Loop Effective Potential

Combining the three contributions (6.17), (6.23), and (6.26), the full one-loop effective potential is

$$V_{\text{eff}}(\bar\varphi) = U(\bar\varphi) + \frac{1}{64\pi^2}\mathrm{Tr}(W^2\ln W) - \frac{4}{64\pi^2}\mathrm{Tr}_f\left[(\bar\varphi\,\Gamma)^4\ln(\bar\varphi\,\Gamma)^2\right] + \frac{3}{64\pi^2}\mathrm{Tr}(M^4\ln M^2)\big|_{\varphi=\bar\varphi}, \tag{6.27}$$

where all traces are evaluated at $\varphi = \bar{\varphi}$. This is the master formula for the one-loop effective potential in a non-Abelian gauge theory. The scalar, fermion, and gauge boson loops each contribute a term of the Coleman-Weinberg form $m^4 \ln m^2$, with the mass matrix appropriate to each field type. In the next chapter we apply this formula to GUT models, where the gauge group is larger and the symmetry breaking pattern is richer.

Chapter 7

The Glashow-Salam-Weinberg Model

The Glashow-Salam-Weinberg (GSW) model is the quantum field theory of electroweak interactions, unifying the electromagnetic and weak forces into a single gauge theory with gauge group $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$. It is one of the cornerstones of the Standard Model and provides the first physical example of spontaneous symmetry breaking in a gauge theory, the Higgs mechanism, which we studied abstractly in the previous chapter. Here we apply the effective potential formalism to this model, computing the field-dependent gauge boson masses and the resulting one-loop correction to the effective potential.

7.1 The Electroweak Lagrangian

The gauge fields of the theory are the $\mathrm{SU}(2)_L$ triplet W^i_μ $(i = 1, 2, 3)$ and the $\mathrm{U}(1)_Y$ singlet V_μ, with corresponding field strengths $W^i_{\mu\nu}$ and $V_{\mu\nu}$. The matter content includes left-handed fermion doublets and right-handed singlets, but for the purposes of computing the effective potential it is the scalar sector that is relevant. The Higgs field φ is a complex doublet under $\mathrm{SU}(2)_L$ with hypercharge $Y = 1$. The Lagrangian is

$$\mathcal{L} = -\frac{1}{4} W^i_{\mu\nu} W^{i\,\mu\nu} - \frac{1}{4} V_{\mu\nu} V^{\mu\nu} + (D_\mu \varphi)^\dagger (D^\mu \varphi) - V(\varphi), \tag{7.1}$$

where the covariant derivative is

$$D_\mu = \partial_\mu - ig \frac{\sigma^i}{2} W^i_\mu - ig' \frac{Y}{2} V_\mu, \tag{7.2}$$

with g and g' the $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ coupling constants respectively, and σ^i the Pauli matrices. The classical potential $V(\varphi) = -\mu^2 \varphi^\dagger \varphi + \lambda(\varphi^\dagger \varphi)^2$ has a minimum at a nonzero field value when $\mu^2 > 0$, triggering spontaneous symmetry breaking.

7.2 Spontaneous Symmetry Breaking and Mass Generation

The Higgs field acquires a vacuum expectation value

$$\langle \varphi \rangle = \begin{pmatrix} 0 \\ a \end{pmatrix}, \tag{7.3}$$

which breaks $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ down to the electromagnetic $\mathrm{U}(1)_{\mathrm{em}}$. The gauge boson masses are read off from the kinetic term evaluated at the VEV. Since the VEV is constant, the ∂_μ term vanishes and

$$D_\mu\langle\varphi\rangle = \left(-ig\frac{\sigma^i}{2}W^i_\mu - ig'\frac{Y}{2}V_\mu\right)\begin{pmatrix}0\\a\end{pmatrix}. \tag{7.4}$$

Expanding the SU(2) part using

$$\sigma^i W^i_\mu = \begin{pmatrix}W^3_\mu & W^1_\mu - iW^2_\mu\\ W^1_\mu + iW^2_\mu & -W^3_\mu\end{pmatrix}, \tag{7.5}$$

acting on the VEV gives

$$\frac{\sigma^i}{2}W^i_\mu\begin{pmatrix}0\\a\end{pmatrix} = \frac{a}{2}\begin{pmatrix}W^1_\mu - iW^2_\mu\\ -W^3_\mu\end{pmatrix}. \tag{7.6}$$

For the U(1) part, with hypercharge $Y = 1$,

$$\frac{Y}{2}V_\mu\begin{pmatrix}0\\a\end{pmatrix} = \frac{a}{2}\begin{pmatrix}0\\V_\mu\end{pmatrix}. \tag{7.7}$$

Combining,

$$D_\mu\langle\varphi\rangle = -ig\cdot\frac{a}{2}\begin{pmatrix}W^1_\mu - iW^2_\mu\\ -W^3_\mu\end{pmatrix} - ig'\cdot\frac{a}{2}\begin{pmatrix}0\\V_\mu\end{pmatrix} = \frac{ia}{2}\begin{pmatrix}gW^1_\mu - igW^2_\mu\\ -gW^3_\mu + g'V_\mu\end{pmatrix}. \tag{7.8}$$

Taking the conjugate,

$$\overline{D_\mu\langle\phi\rangle} = -\frac{ia}{2}\left(gW^1_\mu + igW^2_\mu,\ -gW^3_\mu + g'V_\mu\right), \tag{7.9}$$

so that

$$\overline{D_\mu\langle\phi\rangle}\, D^\mu\langle\phi\rangle = \frac{a^2}{4}\left(gW^1_\mu + igW^2_\mu,\ -gW^3_\mu + g'V_\mu\right)\begin{pmatrix}gW^{1\mu} - igW^{2\mu}\\ -gW^{3\mu} + g'V^\mu\end{pmatrix}. \tag{7.10}$$

Expanding the matrix product,

$$= \frac{a^2}{4}\left[g^2(W^{1\,2}_\mu + W^{2\,2}_\mu) + g^2W^{3\,2}_\mu + g'^2V^2_\mu - 2gg'W^3_\mu V_\mu\right]. \tag{7.11}$$

The fields W^3_μ and V_μ mix, and to identify the physical mass eigenstates we introduce the Weinberg rotation,

$$A_\mu = \sin\theta_W\, W^3_\mu + \cos\theta_W\, V_\mu, \tag{7.12}$$
$$Z_\mu = \cos\theta_W\, W^3_\mu - \sin\theta_W\, V_\mu, \tag{7.13}$$

where $\sin\theta_W = g'/\sqrt{g^2+g'^2}$ and $\cos\theta_W = g/\sqrt{g^2+g'^2}$. Inverting,

$$W^3_\mu = \sin\theta_W\, A_\mu + \cos\theta_W\, Z_\mu, \tag{7.14}$$
$$V_\mu = \cos\theta_W\, A_\mu - \sin\theta_W\, Z_\mu. \tag{7.15}$$

Substituting into each term,

$$
\begin{aligned}
g^2 W_\mu^{3\,2} &= g^2 \sin^2\theta_W A_\mu^2 + g^2\cos^2\theta_W Z_\mu^2 + 2g^2 \sin\theta_W \cos\theta_W A_\mu Z_\mu, \\
g'^2 V_\mu^2 &= g'^2 \cos^2\theta_W A_\mu^2 + g'^2 \sin^2\theta_W Z_\mu^2 - 2g'^2 \sin\theta_W \cos\theta_W A_\mu Z_\mu, \\
-2gg' W_\mu^3 V_\mu &= -2gg'(\sin\theta_W A_\mu + \cos\theta_W Z_\mu)(\cos\theta_W A_\mu - \sin\theta_W Z_\mu) \\
= -2gg'\sin\theta_W\cos\theta_W A_\mu^2 &+ 2gg'\sin^2\theta_W A_\mu Z_\mu - 2gg'\cos^2\theta_W Z_\mu A^\mu + 2gg'\cos\theta_W\sin\theta_W Z_\mu^2.
\end{aligned}
\tag{7.16}
$$

Collecting by field bilinear,

$$
\begin{aligned}
g^2 W_\mu^{3\,2} + g'^2 V_\mu^2 - 2gg' W_\mu^3 V^\mu = {} & A_\mu^2 \left(g\sin\theta_W - g'\cos\theta_W\right)^2 \\
& + Z_\mu^2 \left(g\cos\theta_W + g'\sin\theta_W\right)^2 \\
& + 2\sin\theta_W\cos\theta_W A_\mu Z^\mu \left(g^2 - g'^2 + gg'\left(\frac{\sin\theta_W}{\cos\theta_W} - \frac{\cos\theta_W}{\sin\theta_W}\right)\right).
\end{aligned}
\tag{7.17}
$$

The coefficient of A_μ^2 vanishes identically,

$$
g\sin\theta_W - g'\cos\theta_W = \frac{gg'}{\sqrt{g^2+g'^2}} - \frac{g'g}{\sqrt{g^2+g'^2}} = 0, \tag{7.18}
$$

and the $A_\mu Z^\mu$ cross term also vanishes. The coefficient of Z_μ^2 evaluates to

$$
\left(g\cos\theta_W + g'\sin\theta_W\right)^2 = \left(\frac{g^2}{\sqrt{g^2+g'^2}} + \frac{g'^2}{\sqrt{g^2+g'^2}}\right)^2 = g^2 + g'^2. \tag{7.19}
$$

The photon therefore remains massless, as required by the unbroken $\mathrm{U}(1)_{\mathrm{em}}$, and the kinetic term becomes

$$
\boxed{\overline{D_\mu\langle\varphi\rangle}\, D^\mu\langle\varphi\rangle = \frac{a^2}{4}\left[g^2\left(W_\mu^{1\,2} + W_\mu^{2\,2}\right) + \left(g^2+g'^2\right) Z_\mu^2\right].} \tag{7.20}
$$

Reading off the field-dependent masses,

$$
m^2(W_{1,2}) = \frac{1}{4} g^2 a^2, \tag{7.21}
$$

$$
m^2(Z) = \frac{1}{4}(g^2+g'^2)a^2, \tag{7.22}
$$

while the photon remains massless. These are the standard results of the Higgs mechanism in the electroweak theory.

7.3 One-Loop Gauge Field Contribution to the Effective Potential

With the field-dependent gauge boson masses identified, we can compute their contribution to the one-loop effective potential. By the same logic as in the scalar case, each massive gauge boson with field-dependent mass matrix M^2 contributes a Coleman-Weinberg type term at one loop. The gauge field contribution is

$$V_g^{(1)}(\bar{\varphi}) = \frac{3}{64\pi^2}\,\mathrm{Tr}\big(M^4 \ln M^2\big)\Big|_{\rho=\bar{\varphi}}, \qquad M^2_{\alpha\beta} = \lambda_\alpha\lambda_\beta\,\mathrm{Tr}\big[(t_\alpha\varphi)^\dagger\tau_\beta\varphi\big]\,, \tag{7.23}$$

where the factor of 3 accounts for the three physical polarizations of each massive gauge boson. Substituting the masses of the two W bosons and the Z boson,

$$\begin{aligned} V_g^{(1)}(\bar{\varphi}) &= \frac{3}{64\pi^2}\left(\frac{1}{16}g^4 + \frac{1}{16}g^4 + \frac{1}{16}(g^2+g'^2)^2\right)\bar{\varphi}^4\ln\bar{\varphi}^2 \\ &= \frac{3}{64\pi^2}\cdot\frac{1}{16}\left[2g^4 + (g^2+g'^2)^2\right]\bar{\varphi}^4\ln\bar{\varphi}^2, \end{aligned} \tag{7.24}$$

$$\boxed{\Rightarrow \quad V_g^{(1)}(\bar{\varphi}) = \frac{3}{1024\pi^2}\left[2g^4 + (g^2+g'^2)^2\right]\bar{\varphi}^4\ln\bar{\varphi}^2.} \tag{7.25}$$

The coefficient is determined entirely by the gauge couplings g and g', and the structure $\bar{\varphi}^4\ln\bar{\varphi}^2$ is the characteristic signature of the Coleman-Weinberg mechanism in a gauge theory.

Introduce $g = e/\sin\theta_W$ and $g' = e/\cos\theta_W$ to express the result in terms of measurable quantities. Computing each term,

$$g^4 = \frac{e^4}{\sin^4\theta_W}, \tag{7.26}$$

$$(g^2+g'^2)^2 = \frac{e^4}{\sin^4\theta_W\cos^4\theta_W}. \tag{7.27}$$

Combining,

$$2g^4 + (g^2+g'^2)^2 = \frac{e^4}{\sin^4\theta_W}\left(2 + \frac{1}{\cos^4\theta_W}\right) = \frac{e^4\,(2+\cos^{-4}\theta_W)}{\sin^4\theta_W}. \tag{7.28}$$

The prefactor rewrites as

$$\frac{3}{1024\pi^2} = \frac{3}{64}\cdot\frac{1}{16\pi^2} = \frac{3}{64}\left(\frac{1}{4\pi}\right)^2. \tag{7.29}$$

$$\frac{3}{1024\pi^2}\left[2g^4 + (g^2+g'^2)^2\right] = \frac{3}{64}\left(\frac{e^2}{4\pi}\right)^2\frac{2+\cos^{-4}\theta_W}{\sin^4\theta_W}, \tag{7.30}$$

and therefore

$$V_g^{(1)}(\bar{\varphi}) = \frac{3}{64}\left(\frac{e^2}{4\pi}\right)^2\left[\frac{2+\cos^{-4}\theta_W}{\sin^4\theta_W}\right]\bar{\varphi}^4\ln\bar{\varphi}^2 + \mathcal{O}(\bar{\varphi}^4), \tag{7.31}$$

which expresses the gauge boson contribution entirely in terms of the fine structure constant $\alpha_{\rm em} = e^2/4\pi$ and the Weinberg angle θ_W.

We now derive the renormalized form of the gauge boson contribution. For the GSW model, the field-dependent masses are $m^2(W_{1,2}) = \frac{1}{4}g^2\bar{\varphi}^2$ for the two W bosons and $m^2(Z) = \frac{1}{4}(g^2+g'^2)\bar{\varphi}^2$ for the Z boson. Each massive gauge boson with mass $m^2 = c\bar{\varphi}^2$ contributes to the one-loop integral, with three polarizations, a term

$$\frac{3c^2}{64\pi^2}\bar{\varphi}^4\left[\ln\frac{c\bar{\varphi}^2}{\Lambda^2} - \frac{1}{2}\right]. \tag{7.32}$$

Summing over the two W bosons and the Z boson and defining

$$A \equiv \frac{3}{1024\pi^2}\left[2g^4 + (g^2 + g'^2)^2\right], \tag{7.33}$$

the one-loop gauge boson potential, after the mass renormalization condition has been imposed to cancel the $\Lambda^2\bar{\varphi}^2$ divergence, takes the form

$$V_g^{(1)}(\bar{\varphi}) = C^{(1)}\bar{\varphi}^4 + A\bar{\varphi}^4\left[\ln\frac{\bar{\varphi}^2}{\Lambda^2} - \frac{1}{2}\right]. \tag{7.34}$$

We now compute four derivatives to apply the coupling constant renormalization condition. Differentiating once,

$$\begin{aligned}\frac{d}{d\bar{\varphi}}\left[A\bar{\varphi}^4\left(\ln\frac{\bar{\varphi}^2}{\Lambda^2} - \frac{1}{2}\right)\right] &= 4A\bar{\varphi}^3\left(\ln\frac{\bar{\varphi}^2}{\Lambda^2} - \frac{1}{2}\right) + A\bar{\varphi}^4\cdot\frac{2}{\bar{\varphi}}\\ &= 4A\bar{\varphi}^3\ln\frac{\bar{\varphi}^2}{\Lambda^2} - 2A\bar{\varphi}^3 + 2A\bar{\varphi}^3\\ &= 4A\bar{\varphi}^3\ln\frac{\bar{\varphi}^2}{\Lambda^2},\end{aligned} \tag{7.35}$$

so that

$$\frac{dV_g^{(1)}}{d\bar{\varphi}} = 4C^{(1)}\bar{\varphi}^3 + 4A\bar{\varphi}^3\ln\frac{\bar{\varphi}^2}{\Lambda^2}. \tag{7.36}$$

Differentiating again,

$$\begin{aligned}\frac{d}{d\bar{\varphi}}\left[4A\bar{\varphi}^3\ln\frac{\bar{\varphi}^2}{\Lambda^2}\right] &= 12A\bar{\varphi}^2\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 4A\bar{\varphi}^3\cdot\frac{2}{\bar{\varphi}}\\ &= 12A\bar{\varphi}^2\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 8A\bar{\varphi}^2,\end{aligned} \tag{7.37}$$

$$\frac{d^2V_g^{(1)}}{d\bar{\varphi}^2} = 12C^{(1)}\bar{\varphi}^2 + 12A\bar{\varphi}^2\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 8A\bar{\varphi}^2. \tag{7.38}$$

The third derivative,

$$\begin{aligned}\frac{d}{d\bar{\varphi}}\left[12A\bar{\varphi}^2\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 8A\bar{\varphi}^2\right] &= 24A\bar{\varphi}\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 12A\bar{\varphi}^2\cdot\frac{2}{\bar{\varphi}} + 16A\bar{\varphi}\\ &= 24A\bar{\varphi}\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 24A\bar{\varphi} + 16A\bar{\varphi}\\ &= 24A\bar{\varphi}\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 40A\bar{\varphi},\end{aligned} \tag{7.39}$$

$$\frac{d^3V_g^{(1)}}{d\bar{\varphi}^3} = 24C^{(1)}\bar{\varphi} + 24A\bar{\varphi}\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 40A\bar{\varphi}. \tag{7.40}$$

The fourth derivative,

$$\begin{aligned}\frac{d}{d\bar{\varphi}}\left[24A\bar{\varphi}\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 40A\bar{\varphi}\right] &= 24A\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 24A\bar{\varphi}\cdot\frac{2}{\bar{\varphi}} + 40A\\ &= 24A\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 48A + 40A\\ &= 24A\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 88A,\end{aligned} \tag{7.41}$$

$$\frac{d^4V_g^{(1)}}{d\bar{\varphi}^4} = 24C^{(1)} + 24A\ln\frac{\bar{\varphi}^2}{\Lambda^2} + 88A. \tag{7.42}$$

Imposing the coupling constant renormalization condition $\left.\dfrac{d^4V_g^{(1)}}{d\bar{\varphi}^4}\right|_{\bar{\varphi}=M} = \lambda$,

$$24C^{(1)} + 24A\ln\frac{M^2}{\Lambda^2} + 88A = \lambda, \tag{7.43}$$

and solving for $C^{(1)}$,

$$C^{(1)} = \frac{\lambda - 88A}{24} - A\ln\frac{M^2}{\Lambda^2}. \tag{7.44}$$

Substituting back,

$$\begin{aligned} V_g^{(1)}(\bar{\varphi}) &= \left[\frac{\lambda - 88A}{24} - A\ln\frac{M^2}{\Lambda^2}\right]\bar{\varphi}^4 + A\bar{\varphi}^4\left[\ln\frac{\bar{\varphi}^2}{\Lambda^2} - \frac{1}{2}\right] \\ &= \frac{\lambda}{24}\bar{\varphi}^4 - \frac{88A}{24}\bar{\varphi}^4 - A\bar{\varphi}^4\ln\frac{M^2}{\Lambda^2} + A\bar{\varphi}^4\ln\frac{\bar{\varphi}^2}{\Lambda^2} - \frac{A}{2}\bar{\varphi}^4. \end{aligned} \tag{7.45}$$

The two logarithms combine as

$$A\bar{\varphi}^4\ln\frac{\bar{\varphi}^2}{\Lambda^2} - A\bar{\varphi}^4\ln\frac{M^2}{\Lambda^2} = A\bar{\varphi}^4\ln\frac{\bar{\varphi}^2}{M^2}, \tag{7.46}$$

and the remaining constant terms collect as

$$-\frac{88A}{24} - \frac{A}{2} = -\frac{88A}{24} - \frac{12A}{24} = -\frac{100A}{24} = -\frac{25A}{6}. \tag{7.47}$$

Therefore,

$$V_g^{(1)}(\bar{\varphi}) = \frac{\lambda}{4!}\bar{\varphi}^4 + A\bar{\varphi}^4\left[\ln\frac{\bar{\varphi}^2}{M^2} - \frac{25}{6}\right], \tag{7.48}$$

and substituting back the definition of A and expressing in terms of the measurable quantities e and θ_W via (6.25),

$$\boxed{V_g^{(1)}(\bar{\varphi}) = \frac{\lambda}{4!}\bar{\varphi}^4 + \frac{3}{64}\left(\frac{e^2}{4\pi}\right)^2\left[\frac{2+\cos^{-4}\theta_W}{\sin^4\theta_W}\right]\bar{\varphi}^4\left[\ln\frac{\bar{\varphi}^2}{M^2} - \frac{25}{6}\right].} \tag{7.49}$$

The $-25/6$ arises from collecting the finite constants after renormalization, specifically $-88/24 - 1/2 = -25/6$, and is a universal consequence of the renormalization condition imposed at the scale M, independent of the specific values of the gauge couplings.

7.4 Georgy-Glashow minimal SU(5) model

Chapter 8

Symmetry Breaking in Grand Unified Models

Part II

Finite Temperature Field Theory

Chapter 9

Finite Temperature Field Theory: General formalism

Conventional quantum field theory is formulated in empty space. The LSZ formalism expresses scattering matrix elements in terms of zero-temperature Green's functions, that is, vacuum expectation values of time-ordered strings of Heisenberg field operators. This is perfectly justified for particle accelerator experiments, where the background state is essentially the vacuum. In early universe cosmology, however, the background state is a thermal bath at temperature T, and a different formalism is required.

9.1 Finite-Temperature Green's Functions

The finite-temperature Green's function for a scalar field φ is defined by replacing the vacuum expectation value with a thermal average over the grand canonical ensemble,

$$G_n^\beta(x_1,\ldots,x_n) = \frac{\mathrm{Tr}\left\{e^{-\beta H}\, T\left[\varphi(x_1)\cdots\varphi(x_n)\right]\right\}}{\mathrm{Tr}\, e^{-\beta H}}, \tag{9.1}$$

where $\beta = 1/T$ is the inverse temperature and the trace is taken over all states of the system. This is the average value of the time-ordered product of n field operators in the grand canonical ensemble with no chemical potential.

The finite-temperature generating functional is defined analogously,

$$Z^\beta(J) = \frac{\mathrm{Tr}\left[e^{-\beta H}\, T \exp\!\left(i\int J(x)\,\varphi(x)\, d^4x\right)\right]}{\mathrm{Tr}\, e^{-\beta H}}. \tag{9.2}$$

The essential difference between the zero-temperature and finite-temperature formalisms lies in the boundary conditions on the set of paths over which the functional integral is taken. At zero temperature the transition amplitude is

$${}_H\langle\psi_1,t_1|\psi_2,t_2\rangle_H = \int [d\varphi(s,\mathbf{x})]\,\psi_1^*[\varphi(t_2,\mathbf{x})]\,\psi_2[\varphi(t_1,\mathbf{x})]\,\exp\left[i\int_{t_i}^{t_f} ds\int d^3x\,\mathcal{L}(\varphi,\partial\varphi)\right], \tag{9.3}$$

with $t_i \to -\infty$ and $t_f \to +\infty$. At finite temperature the boundary wavefunctionals $\psi^*[\varphi(t_f,\mathbf{x}_f)]\,\psi[\varphi(t_i,\mathbf{x}_i)]$ are replaced by

$$\sum e^{-\beta E(\psi)}\,\psi^*[\varphi(t_f,\mathbf{x}_f)]\,\psi[\varphi(t_i,\mathbf{x}_i)], \tag{9.4}$$

which implements the thermal weighting of states. It is also worth noting that the finite-temperature effective potential is not gauge invariant: only in physical gauges does the trace correspond to a sum over all physical states of the system.

9.2 Periodicity and the KMS Condition

A fundamental consequence of the thermal boundary conditions is that the finite-temperature Green's functions satisfy periodicity conditions in Euclidean time. Working with Euclidean times $t \in [0, -i\beta]$, the time-ordering is

$$T\left[\varphi(x)\,\varphi(y)\right] = \begin{cases} \varphi(x)\,\varphi(y) & \text{if } ix^0 > iy^0, \\ (-1)^s\,\varphi(y)\,\varphi(x) & \text{if } ix^0 < iy^0, \end{cases} \tag{9.5}$$

where $s = 0$ for bosonic fields and $s = 1$ for fermionic fields. To derive the periodicity condition, we evaluate the two-point function at $x^0 = 0$ with $iy^0 > ix^0$,

$$\begin{aligned} \left(\mathrm{Tr}\,e^{-\beta H}\right) G_2^\beta(x-y)\big|_{x^0=0} &= (-1)^s\,\mathrm{Tr}\left[e^{-\beta H}\,\varphi(y)\,\varphi(x)\right] \\ &= (-1)^s\,\mathrm{Tr}\left[e^{-\beta H}\,\varphi(y^0;\mathbf{y})\,\varphi(0,\mathbf{x})\right] \\ &= (-1)^s\,\mathrm{Tr}\left[\varphi(0,\mathbf{x})\,e^{-\beta H}\,\varphi(y^0;\mathbf{y})\right] \\ &= (-1)^s\,\mathrm{Tr}\left[e^{-\beta H}\,\underbrace{e^{\beta H}\,\varphi(0,\mathbf{x})\,e^{-\beta H}}_{\varphi(-i\beta,\mathbf{x})}\,\varphi(y^0;\mathbf{y})\right] \\ &= (-1)^s\,\mathrm{Tr}\left[e^{-\beta H}\,\varphi(-i\beta,\mathbf{x})\,\varphi(y^0;\mathbf{y})\right] \\ &= (-1)^s\left(\mathrm{Tr}\,e^{-\beta H}\right) G_2^\beta(x-y)\big|_{x^0=-i\beta}, \end{aligned} \tag{9.6}$$

where in the third line we used the cyclicity of the trace, and in the fourth line we used the Heisenberg equation of motion to identify $e^{\beta H}\varphi(0,\mathbf{x})e^{-\beta H} = \varphi(-i\beta,\mathbf{x})$. Dividing through by $\mathrm{Tr}\,e^{-\beta H}$,

$$\boxed{G_2^\beta(x-y)\big|_{x^0=0} = (-1)^s\,G_2^\beta(x-y)\big|_{x^0=-i\beta}.} \tag{9.7}$$

This is the Kubo-Martin-Schwinger (KMS) condition: bosonic Green's functions are periodic in Euclidean time with period β, while fermionic Green's functions are antiperiodic.

9.3 Matsubara Frequencies and Feynman Rules

The periodicity in Euclidean time has a direct consequence for the momentum-space structure of the theory. Vertex factors are unaffected by the change in boundary conditions, and the momentum-space propagator remains of the same form. However, the periodicity forces the energy component of the loop momentum to be discretized. The allowed frequencies, known as Matsubara frequencies, are

$$\omega_n = \frac{2n\pi}{\beta} \qquad \text{(Bose fields)}, \tag{9.8}$$

$$\omega_n = \frac{(2n+1)\pi}{\beta} \qquad \text{(Fermi fields)}, \tag{9.9}$$

for integer n. The Feynman rules are modified accordingly. The correspondence between zero-temperature and finite-temperature rules is summarized in the following table.

	Zero temperature	Finite temperature
Loop integral	$\displaystyle\int \frac{d^4k}{(2\pi)^4}$	$\displaystyle\frac{1}{-i\beta}\sum_{n=-\infty}^{\infty}\int \frac{d^3k}{(2\pi)^3}$
Vertex δ-function	$(2\pi)^4\,\delta^{(4)}(\sum k_i)$	$\displaystyle\frac{\beta}{i}(2\pi)^3\,\delta_{\sum\omega_n}\,\delta^{(3)}\Big(\sum \mathbf{k}_i\Big)$

The replacement of the continuous energy integral by a discrete sum over Matsubara frequencies is the defining feature of the imaginary-time formalism. In the next section we use these rules to compute the finite-temperature corrections to the effective potential.

Chapter 10

Finite-Temperature Corrections to the Effective Potential

10.1 Corrections to Scalar Field Theory

In the zero-temperature case, the one-loop contribution to the effective potential came from the sum

$$i\sum_{n=1}^{\infty}\int\frac{d^4k}{(2\pi)^4}\frac{1}{2n}\left[\frac{\frac{1}{2}\lambda\bar{\varphi}^2}{k^2+i\varepsilon}\right]^n=-\frac{i}{2}\int\frac{d^4k}{(2\pi)^4}\ln\left[1-\frac{\frac{1}{2}\lambda\bar{\varphi}^2}{k^2+i\varepsilon}\right]. \tag{10.1}$$

At finite temperature, the replacement of the continuous energy integral by a discrete Matsubara sum gives

$$V_{\text{eff}}^{(1)\beta}(\bar{\varphi})=U(\bar{\varphi})-\frac{i}{2}\cdot\frac{1}{-i\beta}\sum_{n=-\infty}^{\infty}\int\frac{d^3k}{(2\pi)^3}\ln\left[1-\frac{\lambda\bar{\varphi}^2}{2(\omega_n^2-k^2)}\right], \tag{10.2}$$

$$=U(\bar{\varphi})+\frac{1}{2\beta}\sum_{n=-\infty}^{\infty}\int\frac{d^3k}{(2\pi)^3}\ln\left[1-\frac{\lambda\bar{\varphi}^2}{2(\omega_n^2-k^2)}\right]. \tag{10.3}$$

For Bose fields, the Matsubara frequencies satisfy $-i\beta\omega_n=2\pi n$, giving $\omega_n=2\pi in/\beta$ and $\omega_n^2=-4\pi^2n^2/\beta^2$. Substituting,

$$V_{\text{eff}}^{(1)\beta}(\bar{\varphi})=U(\bar{\varphi})+\frac{1}{2\beta}\sum_{n=-\infty}^{\infty}\int\frac{d^3k}{(2\pi)^3}\ln\left[1+\frac{\lambda\bar{\varphi}^2}{2\left(\frac{4\pi^2}{\beta^2}n^2+k^2\right)}\right]. \tag{10.4}$$

Writing the argument of the logarithm as a ratio,

$$\begin{aligned}V_{\text{eff}}^{(1)\beta}(\bar{\varphi})=U(\bar{\varphi})&+\frac{1}{2\beta}\sum_{n=-\infty}^{\infty}\int\frac{d^3k}{(2\pi)^3}\\&\times\left(\ln\left[\frac{4\pi^2}{\beta^2}n^2+k^2+\frac{\lambda\bar{\varphi}^2}{2}\right]-\ln\left[\frac{4\pi^2}{\beta^2}n^2+k^2\right]\right),\end{aligned} \tag{10.5}$$

where the second term contributes only a field-independent constant and is dropped. Denoting $E_k^2=k^2+\frac{\lambda\bar{\varphi}^2}{2}$,

$$V_{\text{eff}}^{(1)\beta}(\bar{\varphi})=U(\bar{\varphi})+\frac{1}{2\beta}\int\frac{d^3k}{(2\pi)^3}\sum_{n=-\infty}^{\infty}\ln\left[\frac{4\pi^2}{\beta^2}n^2+E_k^2\right]+\text{const.} \tag{10.6}$$

Evaluation of the Matsubara Sum

We evaluate the sum

$$X = \sum_{n=-\infty}^{\infty} \ln\left[\frac{4\pi^2}{\beta^2}n^2 + E_k^2\right] \tag{10.7}$$

by differentiating with respect to E_k. Setting $a^2 = 4\pi^2/\beta^2$ and $y = \beta E_k/2\pi$,

$$\begin{aligned}\frac{dX}{dE_k} &= \sum_{n=-\infty}^{\infty} \frac{2E_k}{a^2n^2 + E_k^2} \\ &= \frac{2}{E_k} + \frac{4}{a}\sum_{n=1}^{\infty}\frac{y}{y^2+n^2},\end{aligned} \tag{10.8}$$

where we separated the $n = 0$ term. Using the identity

$$\sum_{n=1}^{\infty}\frac{y}{y^2+n^2} = -\frac{1}{2y} + \frac{\pi}{2}\coth \pi y = -\frac{1}{2y} + \frac{\pi}{2} + \frac{\pi e^{-2\pi y}}{1-e^{-2\pi y}}, \tag{10.9}$$

$$\begin{aligned}\frac{dX}{dE_k} &= \frac{2}{E_k} + \frac{2\beta}{\pi}\left(-\frac{1}{2y} + \frac{\pi}{2} + \frac{\pi e^{-2\pi y}}{1-e^{-2\pi y}}\right) \\ &= \frac{2}{E_k} - \frac{\beta}{\pi y} + \beta + \frac{2\beta e^{-2\pi y}}{1-e^{-2\pi y}} \\ &= \beta + \frac{2\beta e^{-\beta E_k}}{1-e^{-\beta E_k}},\end{aligned} \tag{10.10}$$

where in the last step we used $y = \beta E_k/2\pi$ so that $\beta/(\pi y) = 2/E_k$, cancelling the first term. Integrating back,

$$X = \int\left(\beta + \frac{2\beta e^{-\beta E_k}}{1-e^{-\beta E_k}}\right)dE_k = \beta E_k + 2\ln\left(1-e^{-\beta E_k}\right). \tag{10.11}$$

Substituting back,

$$\begin{aligned}V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) &= U(\bar{\varphi}) + \frac{1}{2\beta}\int\frac{d^3k}{(2\pi)^3}\left(\beta E_k + 2\ln\left(1-e^{-\beta E_k}\right)\right) \\ &= U(\bar{\varphi}) + \int\frac{d^3k}{(2\pi)^3}\left[\frac{E_k}{2} + \frac{1}{\beta}\ln\left(1-e^{-\beta E_k}\right)\right].\end{aligned} \tag{10.12}$$

The first term inside the integral is the zero-point energy. Using the identity

$$i\int_{-\infty}^{\infty}\frac{dx}{2\pi}\ln(-x^2+y^2-i\varepsilon) = y + \text{const.}, \tag{10.13}$$

and rotating to Euclidean space via $k^0 \to ik_E^0$, $-k_0^2 + E_k^2 = k_E^2 + \frac{\lambda\bar{\varphi}^2}{2}$, the zero-temperature part is recovered and we can write

$$V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) = \underbrace{U(\bar{\varphi}) + \frac{1}{2}\int\frac{d^4k_E}{(2\pi)^4}\ln\left[1+\frac{\lambda\bar{\varphi}^2}{2k_E^2}\right]}_{\text{zero-temperature result}} + \underbrace{\frac{1}{2\pi^2\beta}\int_0^{\infty}dk\,k^2\ln\left(1-e^{-\beta E_k}\right)}_{\text{finite-temperature contribution}}. \tag{10.14}$$

Substituting $x = k\beta$, $dk = dx/\beta$, $E_k = (k^2 + \lambda\bar{\varphi}^2/2)^{1/2}$,

$$\Delta V_{\text{eff}}^{(1)}(\bar{\varphi}) = \frac{1}{2\pi^2\beta^4}\int_0^\infty dx\, x^2 \ln\left[1 - \exp\left[-\left(x^2 + \frac{\lambda\bar{\varphi}^2\beta^2}{2}\right)^{1/2}\right]\right]. \tag{10.15}$$

Defining $y^2 = \lambda\bar{\varphi}^2\beta^2/2$, the finite-temperature correction can be written compactly as

$$V_{\text{eff}}^{(1)}(\bar{\varphi}) = V_{\text{eff}}^{(1)\,T=0}(\bar{\varphi}) + \frac{1}{2\pi^2\beta^4}\, I(y), \tag{10.16}$$

where

$$I(y) = \int_0^\infty dx\, x^2 \ln\left[1 - e^{-(x^2+y^2)^{1/2}}\right]. \tag{10.17}$$

The zero-temperature renormalization counterterms also render this finite-temperature contribution finite, so no additional renormalization is needed.

High-Temperature Expansion

In the high-temperature limit $\bar{\varphi}\beta \ll 1$, we expand $I(y)$ in powers of y^2. Setting $f(y) = \ln(1 - e^{-(x^2+y^2)^{1/2}})$,

$$\begin{aligned} f(y) &= f(0) + y^2 f'(0) + \mathcal{O}(y^4) \\ &= \ln(1 - e^{-x}) + y^2 \left[\frac{e^{-(x^2+y^2)^{1/2}}}{1 - e^{-(x^2+y^2)^{1/2}}} \cdot \frac{1}{2(x^2+y^2)^{1/2}}\right]_{y=0} + \mathcal{O}(y^4) \\ &= \ln(1 - e^{-x}) + \frac{y^2}{2x} \cdot \frac{e^{-x}}{1 - e^{-x}} + \mathcal{O}(y^4). \end{aligned} \tag{10.18}$$

Substituting into the integral,

$$I(y) = \int_0^\infty dx\, x^2 \ln(1 - e^{-x}) + \frac{y^2}{2}\int_0^\infty dx\, \frac{xe^{-x}}{1 - e^{-x}} + \mathcal{O}(y^4). \tag{10.19}$$

For the first term, using $\ln(1 - e^{-x}) = -\sum_{n=1}^\infty e^{-nx}/n$,

$$\int_0^\infty dx\, x^2 \ln(1 - e^{-x}) = -\sum_{n=1}^\infty \frac{1}{n}\int_0^\infty dx\, x^2 e^{-nx}. \tag{10.20}$$

Substituting $y = nx$,

$$\int_0^\infty dx\, x^2 e^{-nx} = \frac{1}{n^3}\int_0^\infty dy\, y^2 e^{-y} = \frac{\Gamma(3)}{n^3} = \frac{2}{n^3}, \tag{10.21}$$

so the first term gives

$$-\sum_{n=1}^\infty \frac{1}{n}\cdot\frac{2}{n^3} = -2\sum_{n=1}^\infty \frac{1}{n^4} = -2\zeta(4) = -2\cdot\frac{\pi^4}{90} = -\frac{\pi^4}{45}. \tag{10.22}$$

For the second term, using $e^{-x}/(1 - e^{-x}) = \sum_{n=0}^\infty e^{-(n+1)x}$,

$$\int_0^\infty dx\, \frac{xe^{-x}}{1 - e^{-x}} = \sum_{n=1}^\infty \int_0^\infty dx\, x\, e^{-nx}. \tag{10.23}$$

Substituting $y = nx$,

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \int_0^{\infty} dy\, y\, e^{-y} = \sum_{n=1}^{\infty} \frac{\Gamma(2)}{n^2} = \zeta(2) = \frac{\pi^2}{6}. \tag{10.24}$$

Combining,

$$\boxed{I(y) = -\frac{\pi^4}{45} + \frac{\pi^2 y^2}{12} + \mathcal{O}(y^4).} \tag{10.25}$$

Substituting back with $y^2 = \lambda\bar{\varphi}^2\beta^2/2$,

$$\begin{aligned} V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) &= V_{\text{eff}}^{(1)\,T=0}(\bar{\varphi}) + \frac{1}{2\pi^2\beta^4}\left(-\frac{\pi^4}{45} + \frac{\pi^2}{12}\cdot\frac{\lambda\bar{\varphi}^2\beta^2}{2} + \mathcal{O}(\bar{\varphi}^4)\right) \\ &= \boxed{V_{\text{eff}}^{(1)\,T=0}(\bar{\varphi}) - \frac{\pi^2}{90\beta^4} + \frac{\lambda\bar{\varphi}^2}{24\beta^2} + \mathcal{O}(\bar{\varphi}^4).} \end{aligned} \tag{10.26}$$

The first correction $-\pi^2/(90\beta^4) = -\pi^2T^4/90$ is the familiar Stefan-Boltzmann term, the free energy density of a massless scalar gas. The second correction $\lambda\bar{\varphi}^2/(24\beta^2) = \lambda T^2\bar{\varphi}^2/24$ is a temperature-dependent mass term, positive and proportional to T^2. This term is what drives symmetry restoration at high temperature: it adds a positive contribution to the effective mass of the field, pushing the minimum of the effective potential back toward $\bar{\varphi} = 0$ as T increases.

10.2 Corrections to the GSW Model

The finite-temperature correction to the effective potential in the GSW model receives contributions from the two W bosons and the Z boson, each with three polarizations. Using the field-dependent masses $m(W_{1,2}) = \frac{1}{2}g\bar{\varphi}$ and $m(Z) = \frac{1}{2}(g^2 + g'^2)^{1/2}\bar{\varphi}$, the correction is

$$\Delta V_{\text{eff}}^{(1)\beta}(\bar{\varphi}) = \frac{3}{2\pi^2\beta^4}\left\{2I\left(\tfrac{1}{2}g\bar{\varphi}\beta\right) + I\left[\tfrac{1}{2}(g^2+g'^2)^{1/2}\bar{\varphi}\beta\right]\right\}. \tag{10.27}$$

Expanding each term using $I(y) = -\pi^4/45 + \pi^2y^2/12 + \mathcal{O}(y^4)$, the W boson contribution gives

$$\begin{aligned} 2I\left(\tfrac{1}{2}g\bar{\varphi}\beta\right) &= 2\left[-\frac{\pi^4}{45} + \frac{\pi^2}{12}\cdot\frac{1}{4}g^2\bar{\varphi}^2\beta^2 + \mathcal{O}(\bar{\varphi}^4)\right] \\ &= -\frac{2\pi^4}{45} + \frac{\pi^2}{24}g^2\beta^2\bar{\varphi}^2 + \mathcal{O}(\bar{\varphi}^4), \end{aligned} \tag{10.28}$$

and the Z boson contribution gives

$$\begin{aligned} I\left[\tfrac{1}{2}(g^2+g'^2)^{1/2}\bar{\varphi}\beta\right] &= -\frac{\pi^4}{45} + \frac{\pi^2}{12}\cdot\frac{1}{4}(g^2+g'^2)\bar{\varphi}^2\beta^2 + \mathcal{O}(\bar{\varphi}^4) \\ &= -\frac{\pi^4}{45} + \frac{\pi^2}{48}(g^2+g'^2)\bar{\varphi}^2\beta^2 + \mathcal{O}(\bar{\varphi}^4). \end{aligned} \tag{10.29}$$

Summing these,

$$
\begin{aligned}
\Delta V_{\text{eff}}^{(1)\beta}(\bar\varphi) &= \frac{3}{2\pi^2\beta^4}\left[-\frac{2\pi^4}{45} + \frac{\pi^2}{24}g^2\beta^2\bar\varphi^2 - \frac{\pi^4}{45} + \frac{\pi^2}{48}(g^2+g'^2)\bar\varphi^2\beta^2 + \mathcal{O}(\bar\varphi^4)\right] \\
&= \frac{3}{2\pi^2\beta^4}\left[-\frac{3\pi^4}{45} + \frac{\pi^2\beta^2\bar\varphi^2}{24}\left(g^2 + \frac{g^2}{2} + \frac{g'^2}{2}\right) + \mathcal{O}(\bar\varphi^4)\right] \\
&= \frac{3}{2\pi^2\beta^4}\left[-\frac{\pi^4}{15} + \frac{\pi^2}{48}\beta^2(3g^2+g'^2)\bar\varphi^2 + \mathcal{O}(\bar\varphi^4)\right] \\
&= -\frac{\pi^2}{10\beta^4} + \frac{1}{32\beta^2}(3g^2+g'^2)\bar\varphi^2 + \mathcal{O}(\bar\varphi^4).
\end{aligned} \tag{10.30}
$$

The full finite-temperature effective potential in the GSW model is therefore

$$
\boxed{V_{\text{eff}}^{(1)\beta}(\bar\varphi) = V_{\text{eff}}^{(1)\,T=0}(\bar\varphi) - \frac{\pi^2}{10\beta^4} + \frac{1}{32\beta^2}(3g^2+g'^2)\bar\varphi^2 + \mathcal{O}(\bar\varphi^4).} \tag{10.31}
$$

As in the scalar case, the $-\pi^2/(10\beta^4)$ term is the Stefan-Boltzmann free energy of the gauge bosons, and the $(3g^2+g'^2)\bar\varphi^2/(32\beta^2)$ term is a positive temperature-dependent mass correction that drives symmetry restoration at high temperature.

10.3 Corrections to the Minimal SU(5) Model

For the minimal SU(5) GUT, the gauge bosons acquire a field-dependent mass $m = \sqrt{25/8}\, g\bar\varphi$ from the Higgs mechanism. The finite-temperature correction to the effective potential is

$$
\begin{aligned}
V_{\text{eff}}^{(1)\beta}(\bar\varphi) &= V_{\text{eff}}^{(1)\,T=0}(\bar\varphi) + \frac{18}{\pi^2\beta^4}\, I\left(\sqrt{\tfrac{25}{8}}\, g\bar\varphi\beta\right) \\
&= V_{\text{eff}}^{(1)\,T=0}(\bar\varphi) + \frac{18}{\pi^2\beta^4}\left[-\frac{\pi^4}{45} + \frac{\pi^2}{12}\cdot\frac{25}{8}\, g^2\bar\varphi^2\beta^2 + \mathcal{O}(\bar\varphi^4)\right] \\
&= V_{\text{eff}}^{(1)\,T=0}(\bar\varphi) - \frac{2\pi^2}{5\beta^4} + \frac{18}{\pi^2\beta^4}\cdot\frac{25}{8}\, g^2\bar\varphi^2\beta^2\cdot\frac{\pi^2}{12} + \mathcal{O}(\bar\varphi^4),
\end{aligned} \tag{10.32}
$$

where the factor of 18 counts the degrees of freedom of the heavy gauge bosons in the SU(5) model. Collecting terms,

$$
\boxed{V_{\text{eff}}^{(1)\beta}(\bar\varphi) = V_{\text{eff}}^{(1)\,T=0}(\bar\varphi) - \frac{2\pi^2}{5\beta^4} + \frac{75}{16\beta^2}\, g^2\bar\varphi^2 + \mathcal{O}(\bar\varphi^4).} \tag{10.33}
$$

The structure is the same as in the scalar and GSW cases: a Stefan-Boltzmann term $-2\pi^2/(5\beta^4)$ from the free energy of the gauge bosons, and a positive temperature-dependent mass term $75g^2\bar\varphi^2/(16\beta^2)$ that drives symmetry restoration. The larger coefficient relative to the GSW case reflects the larger gauge group and the correspondingly greater number of degrees of freedom.

Part III

Cosmological Phase Transitions

Chapter 11

A Brief History of the Early Universe and Cosmological Phase Transitions

11.1 Thermal History of the Early Universe

The universe we observe today emerged from an early hot dense state that has been expanding and cooling ever since. Working backwards in time, the thermal history is punctuated by a sequence of phase transitions, each corresponding to the breaking of a symmetry as the temperature fell below a critical value.

The earliest epoch accessible to our theoretical framework is inflation, a period of accelerated expansion during which the universe expanded by a factor of order 10^{26} over a time of order 10^{-36} to 10^{-32} seconds. At the end of inflation began the reheating phase, during which the universe returned to a thermal state at the reheating temperature T_R through the coupling of the inflaton field ϕ to the Standard Model particles.

Subsequent to reheating, the thermal history proceeds through a series of symmetry breaking transitions. At $T \sim 10^{29}\,\mathrm{K} \sim 10^{16}\,\mathrm{GeV}$, there may have been a GUT phase transition,

$$\mathrm{GUT} \;\longrightarrow\; \mathrm{SU}(2)_L \times \mathrm{U}(1), \tag{11.1}$$

though whether this transition occurred and at what scale depends on the specific GUT model. At $T \sim 10^{15}\,\mathrm{K} \sim 100\,\mathrm{GeV}$, the electroweak symmetry was broken,

$$\mathrm{SU}(2)_L \times \mathrm{U}(1) \;\longrightarrow\; \mathrm{U}_{\mathrm{em}}(1), \tag{11.2}$$

and at $T \sim 2 \cdot 10^{12}\,\mathrm{K} \sim 200\,\mathrm{MeV}$ the QCD confinement transition took place. Each of these transitions is described by a scalar field acting as an order parameter that acquires a vacuum expectation value as the temperature falls through T_c.

11.2 Phase Transitions and the Effective Potential

The nature of a cosmological phase transition is determined by the structure of the finite-temperature effective potential $V_{\mathrm{eff}}(\bar{\varphi}, T)$. We illustrate this with the electroweak phase transition, where the order parameter is the Higgs field h. The tree-level Higgs potential at $T = 0$ is

$$V(h) = -m^2 h^2 + \lambda h^4, \qquad m^2 > 0, \quad \lambda > 0, \quad m^2 \sim (100\,\mathrm{GeV})^2. \tag{11.3}$$

At $T \neq 0$, finite-temperature corrections generate a temperature-dependent mass term,

$$\Delta V(h,T) = cT^2h^2 + \cdots\,, \qquad c > 0, \tag{11.4}$$

so that

$$V(h,T) = V(h) + \Delta V(h,T). \tag{11.5}$$

The positive T^2h^2 term competes with the negative $-m^2h^2$ term and tends to push the minimum back toward $h = 0$ at high temperature, restoring the symmetry.

The character of the transition depends on how the minimum of $V(h,T)$ evolves with temperature. If the minimum moves smoothly and continuously from $h \neq 0$ at low T to $h = 0$ at high T, the transition is *second order* (or a crossover).

If instead a potential barrier develops between the two minima at some temperature, so that the two phases coexist and the field must tunnel through the barrier to reach the new minimum, the transition is *first order*.

The potential barrier in a first-order transition can arise from higher-order terms in the potential. For example, a cubic term $\sim h^3$ generates a barrier, though for the Standard Model Higgs potential alone this is not sufficient to produce a first-order transition. There is also the possibility of *symmetry non-restoration*, which occurs when $c < 0$, so that the finite-temperature correction deepens the broken-symmetry minimum rather than restoring it. This situation has attracted considerable attention in recent years.

11.3 Bubble Nucleation

In a first-order phase transition, the universe finds itself temporarily trapped in the false vacuum, the metastable minimum of $V_{\text{eff}}(\bar{\varphi}, T)$. The field cannot roll classically to the true vacuum because of the potential barrier separating the two phases. Instead, the transition proceeds through the nucleation of bubbles of the true vacuum within the surrounding false vacuum. Initially, regions of space where the field has already tunnelled to the true vacuum appear as isolated bubbles,

which then expand, driven by the pressure difference between the two phases. New bubbles nucleate, and eventually the expanding bubbles overlap and collide, filling all of space with the true vacuum and completing the transition.

This process has two important phenomenological consequences that have attracted significant attention. The first is electroweak baryogenesis, in which the out-of-equilibrium dynamics at the bubble wall provides the conditions necessary to generate the observed baryon asymmetry of the universe. The second is the production of a stochastic gravitational wave background from bubble collisions and their interaction with the primordial plasma. The detailed analysis of both of these consequences requires a quantitative understanding of the bubble nucleation rate and bubble wall dynamics, which we develop in Part **??**.

Chapter 12

The Effective Potential in Cosmology

In Part ?? we developed the effective action formalism and computed the effective potential at zero temperature, using a momentum cutoff to regulate the divergent integrals. Here we revisit this computation in the cosmological context, with two differences. First, we use dimensional regularization and the $\overline{\text{MS}}$ renormalization scheme, which is more systematic and avoids the ambiguities associated with a hard cutoff. Second, we keep the physical application front and center: the question is not just what the quantum-corrected vacuum is, but how it changes as the universe cools, and what determines whether the resulting phase transition is first or second order.

12.1 The Effective Action and Effective Potential

We work with a real scalar field ϕ in ϕ^4 theory,

$$S[\phi] = \int d^4x \left[\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - V_0(\phi)\right], \qquad V_0(\phi) = \frac{m^2}{2}\phi^2 + \frac{\lambda}{4!}\phi^4, \tag{12.1}$$

where the $\phi \to -\phi$ symmetry forbids a cubic term. The basic objects of interest are the correlators

$$\langle\Omega|T\{\phi(x_1)\cdots\phi(x_n)\}|\Omega\rangle = \frac{\int \mathcal{D}\phi\;\phi(x_1)\cdots\phi(x_n)\,e^{iS[\phi]}}{\int \mathcal{D}\phi\;e^{iS[\phi]}}, \tag{12.2}$$

where $|\Omega\rangle$ is the vacuum of the interacting theory. Our goal is to find a potential for the expectation value $\langle\Omega|\phi(x)|\Omega\rangle$.

Introducing a source $J(x)$, the generating functional is

$$Z[J] = \int \mathcal{D}\phi\; e^{iS[\phi]+i\int d^4x\,J(x)\phi(x)}, \tag{12.3}$$

from which all correlators are obtained by functional differentiation. The connected generating functional $W[J] = -i\ln Z[J]$ generates the connected Green's functions. Its first derivative defines the classical field,

$$\frac{\delta W[J]}{\delta J(x)} = \langle\Omega|\phi(x)|\Omega\rangle_J \equiv \bar{\phi}(x). \tag{12.4}$$

The effective action is defined by the Legendre transform,

$$\Gamma[\bar{\phi}] \equiv W[J] - \int d^4x\,J(x)\,\bar{\phi}(x), \tag{12.5}$$

and satisfies $\delta\Gamma/\delta\bar{\phi}(y) = -J(y)$. The vacuum of the theory in the absence of sources is therefore determined by

$$\frac{\delta\Gamma[\bar{\phi}]}{\delta\bar{\phi}} = 0. \tag{12.6}$$

For a spatially uniform, time-independent configuration $\bar{\phi}(x) = \bar{\phi}_c$, the effective action reduces to

$$\Gamma[\bar{\phi}_c] = -\int d^4x\, V_{\text{eff}}(\bar{\phi}_c), \tag{12.7}$$

where $V_{\text{eff}}(\bar{\phi}_c)$ is the effective potential, the quantum-corrected potential whose minimum determines the true ground state of the theory. Without proof, $\Gamma[\bar{\phi}]$ is the generating functional of 1PI correlators,

$$V_{\text{eff}}(\bar{\phi}_c) = -\sum_{n=0}^{\infty} \frac{1}{n!}\, \bar{\phi}_c^n\, \Gamma^{(n)}(p_i = 0). \tag{12.8}$$

12.2 The Coleman-Weinberg Potential via Dimensional Regularization

The one-loop correction to the effective potential is

$$V_1(\phi) = \frac{1}{2}\int \frac{d^4p_E}{(2\pi)^4} \ln\left[p_E^2 + \tilde{m}^2(\phi)\right], \tag{12.9}$$

where

$$\tilde{m}^2(\phi) = \frac{d^2V_0(\phi)}{d\phi^2} = m^2 + \frac{\lambda}{2}\phi^2 \tag{12.10}$$

is the field-dependent mass. This integral is UV divergent and must be regulated. We use dimensional regularization, replacing $d = 4$ by $d = 4 - 2\varepsilon$ and introducing a mass scale μ to keep the action dimensionless,

$$V_1(\phi) = \frac{1}{2}\mu^{4-d} \int \frac{d^dk_E}{(2\pi)^d} \ln\left(p_E^2 + \tilde{m}^2(\phi)\right). \tag{12.11}$$

To evaluate this, we differentiate with respect to $\tilde{m}^2(\phi)$ and use the master formula of dimensional regularization,

$$\int \frac{d^dp_E}{(2\pi)^d} \frac{1}{p_E^2 + m^2} = \frac{1}{(4\pi)^{d/2}} (m^2)^{d/2-1} \Gamma\left(1 - \frac{d}{2}\right), \tag{12.12}$$

to obtain

$$\frac{dV_1}{d\tilde{m}^2} = \frac{1}{2}\mu^{4-d} \cdot \frac{1}{(4\pi)^{d/2}} \left(\tilde{m}^2\right)^{d/2-1} \Gamma\left(1 - \frac{d}{2}\right). \tag{12.13}$$

Integrating back and substituting $d = 4 - 2\varepsilon$,

$$V_1(\phi) = \frac{1}{64\pi^2}\, \tilde{m}^4(\phi) \left[-\frac{1}{\varepsilon} + \gamma_E - \ln 4\pi + \ln\frac{\tilde{m}^2(\phi)}{\mu^2} - \frac{3}{2}\right]. \tag{12.14}$$

In the $\overline{\mathrm{MS}}$ scheme, we absorb the divergent combination $\mathcal{C}_{\mathrm{uv}} = 1/\varepsilon - \gamma_E + \ln 4\pi$ into counterterms, leaving the renormalized one-loop potential

$$\boxed{V_1(\phi) = \frac{1}{64\pi^2}\,\tilde{m}^4(\phi)\left[\ln\frac{\tilde{m}^2(\phi)}{\mu^2} - \frac{3}{2}\right].} \tag{12.15}$$

This result generalizes immediately to other fields. For a U(1) gauge field with field-dependent mass $\tilde{m}_A^2(\phi) = 2g^2|\phi|^2$ arising from spontaneous symmetry breaking,

$$V_1^{(A)}(\phi) = \frac{3}{64\pi^2}\,\tilde{m}_A^4(\phi)\left[\ln\frac{\tilde{m}_A^2(\phi)}{\mu^2} - \frac{5}{6}\right], \tag{12.16}$$

where the factor of 3 counts the physical polarizations and the $-5/6$ (rather than $-3/2$) arises from the vector nature of the field, computed in Landau gauge. For Dirac fermions with Yukawa coupling y and field-dependent mass $m_\psi^2 = y^2\phi^2$,

$$V_1^{(\psi)}(\phi) = -\frac{4}{64\pi^2}\,\tilde{m}_\psi^4(\phi)\left[\ln\frac{\tilde{m}_\psi^2(\phi)}{\mu^2} - \frac{3}{2}\right], \tag{12.17}$$

where the overall minus sign comes from the fermion loop and the factor of 4 counts the spinor degrees of freedom. The general formula, summing over all particles coupling to ϕ, is

$$V_{\mathrm{eff}}(\phi) = V_0(\phi) + \frac{1}{64\pi^2}\sum_i n_i\,\tilde{m}_i^4(\phi)\left[\ln\frac{\tilde{m}_i^2(\phi)}{\mu^2} - c_i\right], \tag{12.18}$$

where $n_i = (-1)^F\cdot$ d.o.f. counts degrees of freedom with a minus sign for fermions, and

$$c_i = \begin{cases} \frac{3}{2} & \text{scalars and fermions,} \\ \frac{5}{6} & \text{gauge bosons.} \end{cases} \tag{12.19}$$

This is the Coleman-Weinberg potential in the $\overline{\mathrm{MS}}$ scheme. It supersedes the cutoff-regulated result of Chapter 4 and will be our working formula for the remainder of this part.

12.3 The Coleman-Weinberg Model

We now apply (12.18) to the Coleman-Weinberg model of massless scalar electrodynamics,

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 + |D_\mu\phi|^2 - \frac{\lambda}{4!}|\phi|^4, \qquad D_\mu = \partial_\mu + igA_\mu. \tag{12.20}$$

There is no mass term in the Lagrangian, and the classical action is scale invariant: under $x \to \lambda x$, $A_\mu \to \lambda^{-1}A_\mu$, $\phi \to \lambda^{-1}\phi$, each term in $\mathcal{L}$ scales as λ^{-4}, leaving $S = \int d^4x\,\mathcal{L}$ invariant.

At the quantum level, scale invariance is broken by the logarithms in (12.18). Writing $\phi = \phi_c + \chi_1 + i\chi_2$ and expanding around the background ϕ_c, the field-dependent masses are

$$\tilde{m}_{\chi_1}^2(\phi_c) = \frac{\lambda}{2}\phi_c^2, \tag{12.21}$$

$$\tilde{m}_{\chi_2}^2(\phi_c) = \frac{\lambda}{6}\phi_c^2, \tag{12.22}$$

$$m_A^2(\phi_c) = 2g^2\phi_c^2. \tag{12.23}$$

In the regime $\lambda^2 \ll g^4$, the scalar loop contributions are negligible and the effective potential is dominated by the gauge boson loop,

$$V_{\text{eff}}(\phi_c) \cong \frac{\lambda}{4!}\phi_c^4 + \frac{3}{16\pi^2}\, g^4\phi_c^4 \left[\ln \frac{2g^2\phi_c^2}{\mu^2} - \frac{5}{6}\right]. \tag{12.24}$$

Redefining the renormalization scale as $\tilde{\mu}^2 = e^{5/6}\mu^2/(2g^2)$, this simplifies to

$$V_{\text{eff}}(\phi_c) = \frac{\lambda}{4!}\phi_c^4 + \frac{3g^4}{16\pi^2}\,\phi_c^4 \ln \frac{\phi_c^2}{\tilde{\mu}^2}. \tag{12.25}$$

Defining an effective quartic coupling

$$\lambda_{\text{eff}}(\phi_c) \equiv \lambda + \frac{g^4}{2\pi^2} \ln \frac{\phi_c^2}{\tilde{\mu}^2}, \tag{12.26}$$

the effective potential takes the suggestive form

$$V_{\text{eff}}(\phi_c) = \frac{\lambda_{\text{eff}}(\phi_c)}{4!}\phi_c^4. \tag{12.27}$$

This is precisely analogous to the running coupling in scattering experiments, where the appropriate scale at which to evaluate the coupling is the energy scale of the process. Here the relevant scale is the field value ϕ_c itself. At $\phi_c = \tilde{\mu}$, the logarithm vanishes and $\lambda_{\text{eff}}(\tilde{\mu}) = \lambda$, so $\tilde{\mu}$ plays the role of an RG scale. Quantum corrections have therefore broken the classical scale invariance and generated a preferred scale, producing a nontrivial minimum of the effective potential.

This is the phenomenon of dimensional transmutation already encountered in Chapter 5: a theory with no classical mass scale develops a preferred scale through radiative corrections. The scale $\tilde{\mu}$ at which λ_{eff} is evaluated plays the same role as the renormalization point M in the cutoff treatment of Chapter 4, and the two approaches give the same physical result in different renormalization schemes.

For the Standard Model Higgs, an analogous but more involved running occurs. The tree-level wine-bottle potential has a minimum at $\sim 100\,\text{GeV}$ with positive λ. However, running λ to higher scales using the Standard Model renormalization group equations, the coupling turns negative at energy scales of order 10^{10} to $10^{12}\,\text{GeV}$, implying that the electroweak vacuum is metastable rather than absolutely stable.

The lifetime of this metastable vacuum is many orders of magnitude larger than the age of the universe, so this poses no immediate phenomenological problem. Nevertheless it raises the deep question of why the measured values of the top quark mass and Higgs mass place us precisely in the metastable region of parameter space. For a detailed analysis, see Buttazzo et al., arXiv:1307.3536.

Chapter 13

Finite-Temperature Effective Potential in Cosmology

We now extend the analysis of the previous chapter to finite temperature. The imaginary-time formalism developed in Chapter 9 provides the framework: at temperature $T = 1/\beta$, the partition function is a path integral over fields periodic in Euclidean time with period β, and the loop integrals are replaced by Matsubara sums. Here we apply this machinery to derive the finite-temperature effective potential in the cosmological context, and use it to analyze the temperature dependence of the phase structure.

13.1 The Imaginary-Time Formalism in Brief

The canonical ensemble in quantum mechanics is described by the density matrix $\rho = Z^{-1}e^{-\beta H}$, with partition function $Z = \mathrm{tr}(e^{-\beta H})$ and ensemble averages

$$\langle \mathcal{O} \rangle = \mathrm{tr}(\rho\, \mathcal{O}) = \frac{1}{Z} \sum_n e^{-\beta E_n} \langle \psi_n | \mathcal{O} | \psi_n \rangle. \tag{13.1}$$

In quantum field theory the trace becomes a path integral over field configurations,

$$Z = \mathrm{tr}(e^{-\beta H}) = \int \mathcal{D}\phi\, \langle \phi | e^{-\beta H} | \phi \rangle. \tag{13.2}$$

To cast this in path integral form, we use Feynman's formula for the transition amplitude,

$$\langle \phi_b | e^{-iH(t_b - t_a)} | \phi_a \rangle = \int_{\phi(t_a)=\phi_a}^{\phi(t_b)=\phi_b} \mathcal{D}\phi\; e^{i \int_{t_a}^{t_b} dt \int d^3x\, \mathcal{L}}, \tag{13.3}$$

and analytically continue to imaginary time $\tau = it$. Setting $t_a = 0$ and $t_b = -i\beta$, identifying $\phi_a = \phi_b$ and integrating over field configurations, the partition function becomes

$$Z = \int_{\phi(0,\mathbf{x})=\phi(\beta,\mathbf{x})} \mathcal{D}\phi\; e^{-\int_0^\beta d\tau \int d^3x\, \mathcal{L}_E}, \tag{13.4}$$

where the Euclidean Lagrangian is

$$\mathcal{L}_E = (\partial_\tau \phi)^2 + (\partial_i \phi)^2 + V(\phi). \tag{13.5}$$

The fields satisfy periodic boundary conditions in Euclidean time, $\phi(0,\mathbf{x}) = \phi(\beta,\mathbf{x})$ for bosons and antiperiodic for fermions. The generating functional at finite temperature is

$$Z^\beta[J] = \int \mathcal{D}\phi \, e^{-\int_0^\beta d\tau \int d^3x \, [\mathcal{L}_E - J\phi]}, \tag{13.6}$$

and the chain from $Z^\beta[J]$ to the finite-temperature effective potential $V^\beta_{\text{eff}}(\bar{\phi}_c)$ follows exactly the same Legendre transform logic as at zero temperature,

$$Z^\beta[J] \xrightarrow{\text{Legendre}} \Gamma^\beta[\bar{\phi}] \xrightarrow{\bar{\phi}=\bar{\phi}_c} V^\beta_{\text{eff}}(\bar{\phi}_c). \tag{13.7}$$

The key difference from zero temperature lies in the Feynman rules. The propagator becomes $-i/(\omega_k^2 + \mathbf{p}^2 + m^2)$ in Euclidean space, the vertex is unchanged at $-i\lambda$, and the loop integral is replaced by a Matsubara sum,

$$\int \frac{d^4p}{(2\pi)^4} \longrightarrow \frac{i}{\beta} \sum_{k=-\infty}^{\infty} \int \frac{d^3p}{(2\pi)^3}, \tag{13.8}$$

with bosonic Matsubara frequencies $\omega_k = 2\pi k \beta^{-1}$ and fermionic frequencies $\omega_k = (2k+1)\pi\beta^{-1}$.

13.2 The Thermal Functions J_B and J_F

Following the same steps as in Chapter 9, the one-loop finite-temperature contribution to the effective potential for a scalar field with field-dependent mass $m^2(\phi)$ is

$$V_{1,\beta}(\phi) = \int \frac{d^3p}{(2\pi)^3} \left[\frac{\omega}{2} + \frac{1}{\beta} \ln(1 - e^{-\beta\omega}) \right], \qquad \omega = \sqrt{\mathbf{p}^2 + m^2(\phi)}. \tag{13.9}$$

The first term is the zero-point energy, which upon Wick rotation reproduces the zero-temperature Coleman-Weinberg result. The second term is the genuinely thermal contribution. After the substitution $x = |\mathbf{p}|\beta$, it takes the form

$$V_{1,\beta}(\phi) = \frac{T^4}{2\pi^2} J_B\left(\frac{m^2(\phi)}{T^2} \right), \tag{13.10}$$

where the bosonic thermal function is defined as

$$J_B(y^2) = \int_0^\infty dx \, x^2 \ln\left(1 - e^{-\sqrt{x^2+y^2}}\right). \tag{13.11}$$

For a U(1) gauge field with field-dependent mass $m_A^2(\phi) = 2g^2|\phi|^2$, the three physical polarizations contribute

$$V^{(A_\mu)}_{1,\beta}(\phi) = \frac{3T^4}{2\pi^2} J_B\left(\frac{m_A^2(\phi)}{T^2} \right). \tag{13.12}$$

For Dirac fermions with field-dependent mass $m_f^2 = y^2\phi^2$, the antiperiodic boundary conditions give fermionic Matsubara frequencies $\omega_k = (2k+1)\pi\beta^{-1}$, and the thermal contribution is

$$V^{(\psi)}_{1,\beta}(\phi) = -\frac{2T^4}{\pi^2} J_F\left(\frac{m_f^2(\phi)}{T^2} \right), \tag{13.13}$$

where the fermionic thermal function is

$$J_F(y^2) = \int_0^\infty dx\, x^2 \ln\Big(1 + e^{-\sqrt{x^2+y^2}}\Big)\,, \tag{13.14}$$

with a plus sign in the exponential reflecting the antiperiodic boundary conditions. The high-temperature expansions of these functions, valid for $y^2 = m^2(\phi)/T^2 \ll 1$, are

$$J_B(y^2) \cong -\frac{\pi^4}{45} + \frac{\pi^2}{12}y^2 - \frac{\pi}{6}|y|^3 - \frac{1}{32}y^4 \ln\frac{y^2}{a_B} + \cdots\,, \tag{13.15}$$

$$J_F(y^2) \cong \frac{7\pi^4}{360} - \frac{\pi^2}{24}y^2 - \frac{1}{32}y^4 \ln\frac{y^2}{a_F} + \cdots\,, \tag{13.16}$$

where $a_B = \pi^2 e^{3/2-2\gamma_E}$ and $a_F = 16\pi^2 e^{3/2-2\gamma_E}$. The leading terms $\mp\pi^4/45$ and $7\pi^4/360$ are the free energy densities of massless bosons and fermions respectively, the Stefan-Boltzmann contributions. The $\pm\pi^2 y^2/12$ terms are the temperature-dependent mass corrections already computed in Chapter 9. The cubic term $-\pi|y|^3/6$ in J_B is new and is absent for fermions: it arises from the bosonic Matsubara zero mode and, as we shall see, plays a crucial role in generating a potential barrier and hence a first-order phase transition.

13.3 The Coleman-Weinberg Model at Finite Temperature

We now bring together the zero-temperature and finite-temperature contributions to obtain the full effective potential of the Coleman-Weinberg model at temperature T. The total effective potential is

$$V_{\text{eff}}(\phi, T) = \underbrace{V_0(\phi) + V_{1,\text{CW}}(\phi)}_{\lambda_{\text{eff}}(\phi)\phi^4/4!} + V_{1,\beta}(\phi), \tag{13.17}$$

where the zero-temperature part was computed in the previous chapter. In the regime $\lambda^2 \ll g^4$, the thermal contribution is dominated by the gauge boson loop,

$$V_{1,\beta}(\phi) \cong \frac{3T^4}{2\pi^2} J_B\left(\frac{2g^2\phi^2}{T^2}\right). \tag{13.18}$$

In the high-temperature approximation $T^2 \gg m_A^2(\phi) = 2g^2\phi^2$, expanding J_B,

$$V_{1,\beta}(\phi) \cong \frac{g^2}{4\pi^2}T^2\phi^2 + \mathcal{O}(\phi^3) + \text{const.}, \tag{13.19}$$

so that the full effective potential becomes

$$V_{\text{eff}}(\phi, T) = \frac{\lambda_{\text{eff}}(\phi)}{4!}\phi^4 + \frac{g^2}{4\pi^2}T^2\phi^2 + \cdots \tag{13.20}$$

The positive $T^2\phi^2$ term drives symmetry restoration at high temperature: it adds a positive mass-squared to the field, pushing the minimum back toward $\phi = 0$. At sufficiently high T the symmetric phase is restored. As the universe cools, the coefficient of ϕ^2 decreases and eventually the broken-symmetry minimum at $\phi \neq 0$ becomes energetically favored.

The cubic term $\sim T|\phi|^3$ from J_B is particularly significant. It generates a potential barrier between the symmetric and broken-symmetry phases that persists down to temperatures below the naive critical temperature, making the transition first order. The height and width of this barrier determine the nucleation rate of bubbles of the broken phase within the symmetric phase, which we analyze in detail in Part **??**.

The temperature at which the two minima become degenerate is the critical temperature T_c, derived in Chapter 9 for the purely quartic case. Below T_c the broken-symmetry phase is the global minimum, and the universe undergoes the phase transition. The order and character of this transition, and its cosmological consequences, depend sensitively on the relative magnitudes of the gauge and scalar couplings, and on the full structure of the effective potential including the cubic term from the thermal bosonic loop.

Chapter 14

Effective Action and Effective Potential on an FLRW Background

Part IV

False Vacuum Decay and Bubble Nucleation

Chapter 15

The Problem of False Vacuum Decay

Chapter 16

Fate of the False Vacuum: Semiclassical Theory

Chapter 17

Bubble Nucleation

Chapter 18

Parameters and Gravitational Wave Signatures of a First-Order Transition

Part V

Quantum Field Theory in Curved Spacetime

Chapter 19

Spacetime Causal Structure

We work in units $\hbar = c = G = 1$ and use the mostly plus metric signature $g_{\mu\nu} = (-,+,+,+)$ throughout this part.

19.1 The Schwarzschild Metric and the Tortoise Coordinate

The Schwarzschild metric outside a spherically symmetric mass distribution is

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2\, d\Omega^2, \qquad r > 2M. \tag{19.1}$$

There is an apparent singularity at $r = 2M$, but this is a coordinate artifact rather than a true curvature singularity. To see this, we introduce the tortoise coordinate r_*, defined by

$$r_* = \int \frac{dr}{1 - \frac{2M}{r}} = r + 2M \ln\left(\frac{r}{2M} - 1\right), \qquad 2M < r < \infty, \quad -\infty < r_* < +\infty. \tag{19.2}$$

The name comes from the fact that an infinite coordinate interval $r_* \to -\infty$ corresponds to approaching the horizon $r \to 2M$, so the horizon recedes to minus infinity in this coordinate. In terms of r_*, the t-r part of the metric becomes

$$-\left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} = \left(1 - \frac{2M}{r}\right)(-dt^2 + dr_*^2), \tag{19.3}$$

which is conformally flat, with an overall factor that vanishes at the horizon but leaves the conformal structure intact. Introducing null coordinates

$$u = t - r_*, \qquad v = t + r_*, \tag{19.4}$$

the metric takes the form

$$ds^2 = -\left(1 - \frac{2M}{r}\right) du\, dv + r^2\, d\Omega^2. \tag{19.5}$$

19.2 Kruskal Coordinates and the Maximal Extension

The singularity at $r = 2M$ in the original coordinates is removed by a further change of coordinates. Any transformation of the form $\tilde{u} = f(u)$, $\tilde{v} = g(v)$ preserves the null structure of the metric. We choose

$$\tilde{u} = -4Me^{-u/4M}, \qquad \tilde{v} = 4Me^{v/4M}, \tag{19.6}$$

which maps $u \in (-\infty, +\infty)$ to $\tilde{u} \in (-\infty, 0)$ and $v \in (-\infty, +\infty)$ to $\tilde{v} \in (0, +\infty)$. In these Kruskal coordinates the metric becomes

$$ds^2 = -\frac{2M}{r(\tilde{u},\tilde{v})}e^{-r(\tilde{u},\tilde{v})/2M}\, d\tilde{u}\, d\tilde{v} + r^2(\tilde{u},\tilde{v})\, d\Omega^2, \tag{19.7}$$

where $r(\tilde{u},\tilde{v})$ is determined implicitly by

$$\tilde{u}\,\tilde{v} = 4M^2\Big(1 - \frac{r}{2M}\Big)\, e^{-r/2M}. \tag{19.8}$$

The metric is now manifestly regular at $r = 2M$: when $r = 2M$ either $\tilde{u}$ or $\tilde{v}$ vanishes, but the prefactor $e^{-r/2M}$ is finite. The true singularity at $r = 0$ corresponds to $\tilde{u}\,\tilde{v} = 4M^2$, a hyperbola in the $(\tilde{u},\tilde{v})$ plane. The singularity is genuine because the curvature invariants diverge there, and geodesics reach it in finite affine parameter.

The Kruskal extension reveals four distinct regions of the maximally extended Schwarzschild spacetime. Region I is the exterior of the black hole, $r > 2M$, where $t =$ const lines converge at spacelike infinity and $r =$ const lines are timelike surfaces. Region II is the black hole interior: once inside, the roles of r and t are exchanged, with $r =$ const becoming spacelike surfaces and an observer being unable to remain at constant r. Regions III and IV are the white hole interior and the second exterior region respectively; they are physically irrelevant for realistic gravitational collapse but are part of the maximal extension.

19.3 The Penrose Diagram

Compactifying the Kruskal coordinates via

$$\bar{u} = \tan^{-1}\frac{\tilde{u}}{4M}, \qquad \bar{v} = \tan^{-1}\frac{\tilde{v}}{4M}, \tag{19.9}$$

maps the entire spacetime into a finite region. This choice is not unique; any monotonic functions would preserve the causal relations. The resulting Penrose diagram represents the full causal structure of the spacetime in a finite picture, with null rays propagating at 45 throughout.

The causal structure encoded in the diagram is physically important. An observer in region I can receive signals from region I and from region IV, but cannot communicate with the interior (region II) or with another exterior observer in region III. The singularity at $r = 0$ is a spacelike surface in regions II, meaning it lies in the future of every observer who crosses the horizon, and cannot be avoided.

The Penrose diagram for 2D Minkowski spacetime is obtained by a similar construction. In null coordinates $u = t - x$, $v = t + x$, the metric is $ds^2 = -du\, dv$. The compactification

$$u' = 2\tan^{-1} u, \qquad v' = 2\tan^{-1} v, \qquad -\pi \le u', v' \le \pi, \tag{19.10}$$

maps the full Minkowski space to a finite diamond. The conformal factor is

$$\Omega^2 = \frac{1}{4}\sec^2\frac{u'}{2}\sec^2\frac{v'}{2}, \tag{19.11}$$

and after a conformal rescaling $g_{\mu\nu} \to \Omega^{-2} g_{\mu\nu}$ the metric takes the same form $d\tilde{s}^2 = du'\,dv'$ over the compact region. The boundaries of the diagram correspond to future and past null infinity $I^{\pm}$, future and past timelike infinity $i^{\pm}$, and spacelike infinity i^0. Null rays terminate on $I^{\pm}$ because $u = \tan(u'/2) \to \pm\infty$ as $u' \to \pm\pi$. In 4D the same construction applies, with each interior point representing a 2-sphere, except on the symmetry axis and at i^0.

Appendix A

Proof that $Z(J) = e^{iW(J)}$

The connection between $Z(J)$ and $W(J)$ follows from the combinatorics of Wick diagrams. The S-matrix is

$$S = T \exp \left[-i \int_{-\infty}^{\infty} dt \, H_I(t) \right], \tag{A.1}$$

where H_I is the interaction picture Hamiltonian, which here includes a source term. Expanding S in a Taylor series and applying Wick's theorem gives a sum over Wick diagrams,

$$S = \sum \mathcal{W}, \tag{A.2}$$

where each $\mathcal{W}$ is a Wick graph. For example, for an interaction of the form $H_I(x) = \lambda\bar{\psi}(x)\psi(x)\varphi(x)$, one of the terms at order λ^2 looks like

Unlike Feynman diagrams, Wick diagrams are labelled. Taking the vacuum expectation value of S selects only diagrams with no uncontracted external legs,

$$Z(J) = \langle 0|S|0\rangle = \sum \mathcal{W}_u, \tag{A.3}$$

where $\mathcal{W}_u$ denotes vacuum graphs. A vacuum graph is defined as a graph with no external legs, only internal lines forming closed loops,

Any collection of vacuum graphs combined into a single disconnected diagram is still a vacuum graph, since no external legs appear anywhere. Similarly, $iW(J) = \sum \mathcal{W}_{cu}$, where $\mathcal{W}_{cu}$ are the connected vacuum graphs. It then follows from the theorem proved in Appendix B that

$$Z(J) = \sum \mathcal{W}_u = \exp \sum \mathcal{W}_{cu} = e^{iW(J)}, \tag{A.4}$$

which completes the proof. $\square$

Appendix B

The Sum of All Wick Diagrams is the Exponential of the Connected Ones

We prove that

$$\sum \text{ all Wick diagrams} = \exp\Big(\sum \text{ connected Wick diagrams}\Big). \tag{B.1}$$

Consider the case where H_I contains a single interaction term. Two diagrams belong to the same pattern P if they differ only by a permutation of vertices. Define $S(P)$ as the number of permutations that leave a diagram of pattern P unchanged, and $n(P)$ as the number of vertices. Then the S-matrix is

$$S = \sum_P \frac{:\mathcal{O}(P):}{S(P)}, \tag{1}$$

where $:\mathcal{O}(P):$ is the normal-ordered operator represented by the Wick diagram. Now consider a pattern consisting of n_i copies of the connected pattern P_i, with $i = 1, 2, \ldots$ an ordering of all connected patterns. Then

$$\mathcal{O}(P) = \prod_{i=1}^{\infty} \mathcal{O}(P_i)^{n_i}, \tag{2}$$

$$S(P) = \prod_{i=1}^{\infty} S(P_i)^{n_i}. \tag{3}$$

Substituting into (1), the additional $n_i!$ in the denominator accounts for the fact that permuting graphs of a given connected pattern does not yield a new Wick graph,

$$S = \sum_{\substack{n_i=0 \\ i=0}}^{\infty} \frac{:\prod_{i=1}^{\infty} \mathcal{O}(P_i)^{n_i}:}{\prod_{i=1}^{\infty} [S(P_i)^{n_i}\, n_i!]}. \tag{4}$$

Factoring the sum over each n_i independently,

$$
\begin{aligned}
S &= \prod_{i=1}^{\infty} \sum_{n_i=0}^{\infty} \frac{1}{n_i!} \left[\frac{:\mathcal{O}(P_i):}{S(P_i)} \right]^{n_i} \\
&=: \prod_{i=1}^{\infty} \exp\left(\frac{\mathcal{O}(P_i)}{S(P_i)} \right) : \\
&=: \exp\left(\sum_{i=1}^{\infty} \frac{\mathcal{O}(P_i)}{S(P_i)} \right) : \\
&=: \exp \sum \mathcal{W}_c :, \qquad (5)
\end{aligned}
$$

where $\mathcal{W}_c$ denotes connected Wick graphs and we used (2) and (3) in the last step. This establishes that

$$\sum \text{ all Wick diagrams} = e^{\sum \text{ connected Wick diagrams}}. \qquad (6)$$

$\square$

The proof of $Z(J) = e^{iW(J)}$ in Appendix A then follows immediately by taking the vacuum expectation value of (6).

Appendix C

Equivalence of the Loop Expansion and the $\hbar$ Expansion

We show that an L-loop graph carries a factor of $\hbar^{L-1}$, so that the loop expansion and the $\hbar$ expansion are equivalent. Starting from

$$Z(J) = N \int [\mathcal{D}\varphi] \exp\left[\frac{i}{\hbar} S(\varphi, J)\right], \tag{C.1}$$

the vertex factors are the coefficients of the interaction terms in $\hbar^{-1}S(\varphi, J)$, so each vertex contributes a factor of $\hbar^{-1}$. The propagators are the inverses of the quadratic operators in the exponent, so each internal line contributes a factor of $\hbar$. A graph with V vertices and I internal lines therefore carries an overall factor of

$$\hbar^{I-V}. \tag{C.2}$$

For a connected graph, each vertex fixes one internal momentum except for one overall energy-momentum conservation constraint, so the number of independent loop momenta is

$$L = I - V + 1. \tag{C.3}$$

Hence the $\hbar$-dependence of a connected graph is

$$\hbar^{I-V} = \hbar^{L-1}, \tag{C.4}$$

which shows that the loop expansion is precisely the expansion in powers of $\hbar$. □

Appendix D

Derivation of $\delta W(J)/\delta J = \varphi$

This appendix supplies the step used in the proof of Theorem 3, following Jackiw and Kerman (1979). The two equivalent expressions for $W(J)$ are

$$e^{\frac{i}{\hbar}W(J)} = \langle -, t | +, t \rangle, \tag{D.1}$$

$$e^{\frac{i}{\hbar}W(J)} = \langle 0 | \, T \exp\left(\frac{i}{\hbar} \int (dx)\, J(x)\, \Phi(x) \right) |0\rangle. \tag{D.2}$$

The states $|\pm, t\rangle$ solve the time-dependent Schrödinger equation

$$i\hbar\, \partial_t |\pm, t\rangle = H|\pm, t\rangle - \int d\mathbf{x}\, J(t, \mathbf{x})\, \Phi(\mathbf{x}) |\pm, t\rangle, \tag{D.3}$$

with boundary conditions $\lim_{t\to\mp\infty} |\pm, t\rangle = |0\rangle$, where $|0\rangle$ is the ground state of H with $H|0\rangle = 0$. The states $|\psi_\pm, t\rangle$ are related to $|\pm, t\rangle$ by

$$|+, t\rangle = \exp \frac{i}{\hbar} \left(\int_{-\infty}^{t} dt'\, w(t') \right) |\psi_+, t\rangle, \tag{D.4}$$

$$|-, t\rangle = \exp \frac{i}{\hbar} \left(- \int_{t}^{\infty} dt'\, w^*(t') \right) |\psi_-, t\rangle, \tag{D.5}$$

and satisfy $\big(i\hbar\, \partial_t - H + \int J\Phi\, dx\big)\, |\pm, t\rangle = 0$ with the same boundary conditions. Their overlap gives

$$\langle -, t | +, t \rangle = e^{\frac{i}{\hbar}W(J)}. \tag{D.6}$$

Expanding this explicitly,

$$\begin{aligned}
&\langle \psi_-, t| \left[\exp \frac{i}{\hbar} \left(- \int_{t}^{\infty} dt'\, w^*(t') \right) \right]^* \exp \frac{i}{\hbar} \left(\int_{-\infty}^{t} w(t')\, dt' \right) |\psi_+, t\rangle \\
&= \langle \psi_-, t| \exp \frac{i}{\hbar} \left(\int_{t}^{\infty} dt'\, w(t') + \int_{-\infty}^{t} dt'\, w(t') \right) |\psi_+, t\rangle \\
&= \langle \psi_-, t| \exp \frac{i}{\hbar} \int_{-\infty}^{\infty} dt'\, w(t')\, |\psi_+, t\rangle \\
&= \exp \frac{i}{\hbar} \int_{-\infty}^{\infty} dt'\, w(t')\, \underbrace{\langle \psi_-, t | \psi_+, t\rangle}_{1} \\
&= \exp \frac{i}{\hbar} \left(\int_{-\infty}^{\infty} dt'\, w(t') \right),
\end{aligned} \tag{D.7}$$

so that $W = \int_{-\infty}^{\infty} dt\, w(t)$. Using the eigenvalue equation for $|\psi_+, t\rangle$,

$$\int_{-\infty}^{\infty} dt\, \langle\psi_-, t| \left(i\hbar\,\partial_t - H + \int d\mathbf{x}\, J\Phi\right) |\psi_+, t\rangle = \int_{-\infty}^{\infty} dt\, w(t)\, \underbrace{\langle\psi_-, t|\psi_+, t\rangle}_{1}, \tag{D.8}$$

which gives

$$W(J) = \int_{-\infty}^{\infty} dt\, \langle\psi_-, t|\, i\hbar\,\partial_t - H + \int d\mathbf{x}\, J\Phi\, |\psi_+, t\rangle. \tag{D.9}$$

Taking the functional derivative,

$$\begin{aligned}\frac{\delta W(J)}{\delta J} &= \int_{-\infty}^{\infty} dt\, \frac{\delta\langle\psi_-, t|}{\delta J} \underbrace{\left(i\hbar\,\partial_t - H + \int J\Phi\right) |\psi_+, t\rangle}_{w(t)|\psi_+,t\rangle} \\ &+ \int_{-\infty}^{\infty} dt\, \langle\psi_-, t|\frac{\delta}{\delta J} \left(i\hbar\,\partial_t - H + \int d\mathbf{x}\, J\Phi\right) |\psi_+, t\rangle \\ &+ \int_{-\infty}^{\infty} dt\, \langle\psi_-, t| \left(i\hbar\,\partial_t - H + \int d\mathbf{x}\, J\Phi\right) \frac{\delta|\psi_+, t\rangle}{\delta J},\end{aligned} \tag{D.10}$$

where the constraint $\langle\psi_-, t|\Phi(\mathbf{x})|\psi_+, t\rangle = \phi(t, \mathbf{x})$ was used for the second term. Integrating the third term by parts,

$$i\hbar \left\langle \psi_-, t\middle|\frac{\delta|\psi_+, t\rangle}{\delta J}\right\rangle\Bigg|_{-\infty}^{\infty} + \left\langle \psi_-, t \middle| \underbrace{\left(-i\hbar\,\partial_t - H + \int d\mathbf{x}\, J\Phi\right)}_{\langle\psi_-,t|w(t)} \frac{\delta|\psi_+, t\rangle}{\delta J} \right\rangle, \tag{D.11}$$

so that

$$\begin{aligned}\frac{\delta W(J)}{\delta J} &= \phi + i\hbar \left\langle \psi_-, t\middle|\frac{\delta}{\delta J}|\psi_+, t\rangle\right\rangle\Bigg|_{-\infty}^{\infty} \\ &+ \int_{-\infty}^{\infty} dt\, w(t) \left[\frac{\delta\langle\psi_-, t|}{\delta J}|\psi_+, t\rangle + \langle\psi_-, t|\frac{\delta}{\delta J}|\psi_+, t\rangle\right],\end{aligned} \tag{D.12}$$

where the expression inside the brackets vanishes upon differentiating the normalization constraint $\langle\psi_-, t|\psi_+, t\rangle = 1$. It remains to show that the boundary term also vanishes. At $t \to -\infty$, $|\psi_+, t\rangle \to |0\rangle$ independently of J, so $\delta|\psi_+, t\rangle/\delta J = 0$ at the lower limit. For the upper limit, rewrite it as

$$i\hbar\,\frac{\delta}{\delta J} \underbrace{\langle\psi_-, t|\psi_+, t\rangle}_{1} - i\hbar \left(\frac{\delta\langle\psi_-, t|}{\delta J}\right) |\psi_+, t\rangle, \tag{D.13}$$

where the second term vanishes because at $t \to \infty$, $\langle\psi_-, t| \to \langle 0|$, also independently of J. The boundary term therefore vanishes at both limits, and

$$\therefore\quad \boxed{\frac{\delta W(J)}{\delta J} = \phi.} \tag{D.14}$$

Appendix E

Matrix Element of the Inverse Propagator

We evaluate $\langle k'|\, i\Delta^{-1}[\rho]\,|k\rangle$ using $\langle x|k\rangle = e^{ikx}$,

$$\begin{aligned}
\langle k'|\, i\Delta^{-1}[\rho]\,|k\rangle &= \int d^4x\, d^4y\ \langle k'|x\rangle\langle x|\, i\Delta^{-1}[\rho]\,|y\rangle\langle y|k\rangle \\
&= \int d^4x\, d^4y\ e^{ik'x}\,[-\,(\Box + V''(\rho))]\,\delta^4(x-y)\,e^{iky} \\
&= \int d^4x\ e^{-i(k-k')x}\,\left[-(-k^2 + V''(\rho))\right] \\
&= \left[k^2 - V''(\rho)\right](2\pi)^4\delta^4(k-k'). \qquad \text{(E.1)}
\end{aligned}$$

Setting $k = k'$ and dividing by $\Omega = (2\pi)^4\delta^4(0)$,

$$\langle k|\, i\Delta^{-1}(\rho)\,|k\rangle = \left[k^2 - V''(\rho)\right]\Omega, \qquad \text{(E.2)}$$

$$\Rightarrow \quad \Omega^{-1}\langle k|\, i\Delta^{-1}(\rho)\,|k\rangle = k^2 - V''(\rho). \qquad \text{(E.3)}$$

Appendix F

Evaluation of the One-Loop Momentum Integral

We evaluate

$$I = \int \frac{d^4k_E}{(2\pi)^4} \ln(k_E^2 + V'') = \frac{1}{8\pi^2} \int_0^\Lambda dk_E\, k_E^3 \ln(k_E^2 + V''), \tag{F.1}$$

where the angular integration in four-dimensional Euclidean space gives a factor $2\pi^2$. Substituting $u = k_E^2$, $du = 2k_E\, dk_E$,

$$I = \frac{1}{16\pi^2} \int_0^{\Lambda^2} u\, du \ln(u + V''). \tag{F.2}$$

Integrating by parts with $f = \ln(u + V'')$, $dg = u\, du$,

$$I = \frac{1}{16\pi^2} \left\{ \left[\frac{u^2}{2} \ln(u + V'') \right]_0^{\Lambda^2} - \int_0^{\Lambda^2} \frac{u^2}{2} \cdot \frac{1}{u + V''}\, du \right\}, \tag{F.3}$$

where the boundary term evaluates to $\dfrac{\Lambda^4}{2} \ln(\Lambda^2 + V'')$. Using the partial fraction decomposition $\dfrac{u^2}{u + V''} = u - V'' + \dfrac{(V'')^2}{u + V''}$,

$$\begin{aligned} \frac{1}{2} \int_0^{\Lambda^2} \frac{u^2}{u + V''}\, du &= \frac{1}{2} \int_0^{\Lambda^2} \left(u - V'' + \frac{(V'')^2}{u + V''} \right) du \\ &= \frac{1}{2} \left[\frac{\Lambda^4}{2} - \Lambda^2 V'' + (V'')^2 \ln(\Lambda^2 + V'') - (V'')^2 \ln V'' \right]. \end{aligned} \tag{F.4}$$

Assembling the pieces,

$$I = \frac{1}{16\pi^2} \left\{ \frac{\Lambda^4}{2} \ln(\Lambda^2 + V'') - \frac{1}{2} \left[\frac{\Lambda^4}{2} - \Lambda^2 V'' + (V'')^2 \ln\left(\frac{\Lambda^2 + V''}{V''} \right) \right] \right\}. \tag{F.5}$$

Expanding $\ln(\Lambda^2 + V'') = \ln \Lambda^2 + \ln(1 + V''/\Lambda^2) \simeq \ln \Lambda^2 + V''/\Lambda^2 - (V'')^2/\Lambda^4 + \cdots$, the leading terms are

$$\frac{\Lambda^4}{2} \ln(\Lambda^2 + V'') \simeq \Lambda^4 \ln \Lambda + \frac{\Lambda^2 V''}{2} - \frac{(V'')^2}{2}, \tag{F.6}$$

$$\frac{(V'')^2}{2} \ln(\Lambda^2 + V'') \simeq \frac{(V'')^2}{2} \ln \Lambda^2 + \mathcal{O}(1/\Lambda^2). \tag{F.7}$$

Collecting all terms,

$$I \quad = \quad \frac{1}{16\pi^2}\left\{\Lambda^4\left(\ln\Lambda - \frac{1}{4}\right) + \Lambda^2 V'' + \frac{(V'')^2}{2}\left(\ln\frac{\Lambda^2}{V''} - \frac{1}{2}\right)\right\} \; + \; \mathcal{O}\left(\frac{1}{\Lambda^2}\right). \quad \text{(F.8)}$$

Dropping the $\Lambda^n \ln\Lambda$ terms, which are field-independent and do not affect the location of the minimum of $U(\rho)$, the one-loop integral contributing to (4.49) is

$$\frac{\hbar}{2}I \quad = \quad \hbar\left[\frac{\Lambda^2}{32\pi^2}V''(\rho) + \frac{[V''(\rho)]^2}{64\pi^2}\left(-\frac{1}{2} + \ln\frac{V''(\rho)}{\Lambda^2}\right) + \mathcal{O}\left(\frac{1}{\Lambda^2}\right)\right]. \quad \text{(F.9)}$$

Bibliography